COSMIC THEORY OF CONSCIOUSNESS AND UNIVERSE INHERITANCE

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Maciej Łuczewski

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This theory is the foundation of the science-fiction trilogy "L3" (The Trojan Observer · HORIZON · THE GARDENERS), available on Amazon. Share this file with anyone whose eyes it might open.

Contact: maciejweb@gmail.com · Author website: [address to be added]

COSMIC THEORY OF CONSCIOUSNESS AND UNIVERSE INHERITANCE
A scientific synthesis
The mechanism of multiverse evolution, consciousness as quantum information, and the controlled phase transition of the vacuum
Maciej Łuczewski
May 2026
Version 5.0 — the vacuum as the code of the cosmos, the third path: Mitotic Saturation (an extension of version 4.0 with the reproductive division of universes without destruction)
"The most beautiful experience we can have is the mysterious. It is the fundamental emotion which stands at the cradle of true art and true science." — Albert Einstein
"The most important question a human being can ask is: Is the universe a friendly place?" — Albert Einstein

ABSTRACT

The Cosmic Theory of Consciousness and Universe Inheritance (CTCUI) is a cosmological hypothesis proposing a unified framework for a set of open problems in contemporary physics: the Hubble tension, the nature of dark energy and dark matter, the black hole information paradox, the puzzle of the fine-tuning of the fundamental constants, and the hard problem of consciousness.
At the heart of the theory lie three interconnected theses. First, the universe is not a solitary entity but a node in a generative multiverse population, in which new universes arise from black hole singularities (an extension of Lee Smolin's hypothesis of cosmological natural selection) or from a controlled phase transition of the quantum vacuum from a false to a true minimum. Second, consciousness is quantum information encoded in the quantum zero-point field (ZPF) — not an epiphenomenon of neuronal activity, but a fundamental excitation of the field whose correlations survive the death of the biological carrier and integrate into the deeper structure of reality. Third, the evolution of the multiverse is an informational process: each new universe inherits both the physical parameters and the accumulated quantum information of consciousness from the previous cycle, which gradually optimizes the conditions for the emergence of complex structures and of consciousness itself.
The present synthesis combines the original manifesto of the theory (May 2025) with its mechanistic development, "the Transition Protocol" (March 2026), and with two independent scientific lines emerging in 2025–2026: the work of Slotine and Lohmiller (MIT) demonstrating that the Schrödinger equation can be exactly reproduced by deterministic classical trajectories weighted by density — which provides a mathematical bridge between the coherent structure of information and quantum tunneling of the vacuum — and the LHCb experimental result indicating a 4σ deviation from the Standard Model in penguin decays of B mesons, opening space for new physics coupling to the Higgs field. The work of Joachim Keppler (2025) on the resonance of consciousness with ZPF modes provides a complementary microphysical pillar.
The theory proposes three equally admissible final scenarios: the Controlled Phase Transition (the intentional initiation of a vacuum phase transition by a sufficiently advanced civilization — cosmic genetic engineering at the level of universes), Natural Saturation (the gradual stabilization of the false minimum through the accumulation of coherent consciousness-information — the Universe "grows into" its shelf instead of falling off it), and Mitotic Saturation (the reproductive division of the universe at the saturation threshold, in which the old cosmos endures and a new one arises alongside it, equipped with learned information — without the destruction of the parent). The choice between them has not been settled empirically, but it confronts every sufficiently conscious civilization with a moral dilemma of a fundamental nature, whose shape the third path significantly changes.
CTCUI is a speculative but falsifiable hypothesis. Seven groups of observational predictions have been formulated, covering: (1) the resolution of the metastability of the electroweak vacuum

through more precise measurements of the top quark and Higgs boson masses; (2) correlations between the density of complex structures and local dark energy anomalies; (3) the topology of the nodes of the cosmic web as privileged sites for the phase transition; (4) experimental tests of the Penrose–Hameroff hypothesis on the quantum nature of consciousness; (5) tests of quantum information conservation in processes of the electron–positron annihilation type (Vopson); (6) precise mappings of a possible drift of physical constants in cosmological time as a trace of "teleological" tension in the vacuum field; (7) reproductive signatures in observational data (non-Gaussian CMB inhomogeneities, homologies of constants across generations of universes, anomalies in the Hawking spectrum, multimodal evolution of dark energy, topological traces of earlier acts of division) as empirical tests of the Mitotic Saturation scenario.

Version 3.0 of this synthesis introduced a layer of mathematical operationalization (Part VII). The basic moves included: (a) a reformulation of Keppler's hypothesis into the language of standard quantum electrodynamics (an effective dielectric susceptibility $\chi(\omega)$ of neural tissue instead of the metaphor of "resonance of consciousness with the zero-point field"); (b) the introduction of the consciousness-information functional $\mathcal{J}(t)$ as a measure of ordered correlations in the vacuum field, with an operational definition of the Saturation threshold as the critical value $\mathcal{J}_c$; (c) a simple phenomenological ansatz for dark energy, $\Lambda_{eff}(t) = \Lambda_0 + \alpha \cdot n_{BH}(t) + \beta \cdot \mathcal{J}(t)$, casting the theses of CTCUI into the language of the parameterizations currently used by DESI and Planck; (d) concrete order-of-magnitude targets for dynamical dark energy $w(z)$ and for the drift of the fine-structure constant α. Part VIII introduced a response to the consolation objection in the form of three epistemic moves.

Version 4.0 introduced a central methodological correction in Chapter 23 — an inversion of the conceptual frame that had been missing from all earlier formulations. Version 3.0 had proposed a materialist calculation of the biological contribution to the vacuum energy ($N_{civ} \times N_{brain} \times \delta\rho_{brain}$), which yielded a value of the order of 10^{-65} and was meant to constitute the theory's "first number." That calculation, however, was methodologically inconsistent with CTCUI itself: it treated the brain as an isolated generator of consciousness, whereas the theory has held from its first chapter that consciousness is a fundamental property of the field, and biology is its resonant instrument, not its source. Version 4.0 rejects the materialist frame and introduces an inversion in its place: the zero-point field carries the potential of consciousness (of the order of the ZPF energy, $\sim 10^{120}$ times greater than the observed dark energy), while biology — together with the whole richness of relationships, networks, language, culture, and interspecies contact — *actualizes* a fraction of that potential. Consciousness is measured not by how much biology produces, but by η — the fraction of the available potential that becomes resonantly actualized by the biological network. The functional $\mathcal{J}(t)$ is reformulated within this new frame. The network of relationships, not the isolated brain, becomes the fundamental unit of measurement of consciousness.

Version 5.0 introduces the vacuum-as-code thesis and the Mitotic Saturation scenario as the third path of the end and rebirth of the universe (new Chapter 15 in Part V). The central thesis: the vacuum state with its entire structure of fluctuations is the structural code of the present universe; the vacuum phase transition, triggered by a saturation impulse, is an act of reproductive change of that code, generating a descendant universe *alongside* the old one, not at its expense. The third path eliminates the necessity of choosing between destruction (the Controlled Phase Transition) and stagnation (Natural Saturation): mitotic division preserves the parent and equips the descendant with learned information about which parameters favor complexity and consciousness. The parameter η from Chapter 23 gains an additional interpretation as the degree of epigenetic modulation of the expression of the vacuum code. A seventh group of observational predictions is introduced (Chapter 18.7), covering reproductive signatures: non-Gaussian CMB inhomogeneities, homologies of the fundamental constants, anomalies in the Hawking spectrum, multimodal evolution of $w(z)$, topological traces of earlier divisions. Six open problems — the mechanism of local bubble stabilization, epigenetic signaling on the cosmic scale, the topological parent–descendant coupling, informational limits of inheritance, the relation between the paths, and a full theory of the measurement of η — remain as the research program for the next stage of CTCUI's development.

The article is an interdisciplinary synthesis, based on the published literature of theoretical physics, cosmology, philosophy of mind, and quantum information theory. The author is an independent researcher (amateur scientist in the honest sense of the word); the text does not claim the status of professional work, but rather aims to organize a coherent hypothesis in such a way that it can be taken up, criticized, or developed by professional physicists.

Keywords: multiverse, cosmological natural selection, quantum vacuum, false vacuum, quantum tunneling, dark energy, information paradox, quantum consciousness, integrated information, zero-point field, holographic principle, cosmic web, mitotic saturation, vacuum code, inheritance of universes.

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Consciousness, IIT, quantum biology
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WORK INFORMATION

INTRODUCTION: THREE CRISES AND ONE HYPOTHESIS

Contemporary physics finds itself in a state of deep, though not often openly declared, uncertainty. Three fundamental theoretical frameworks — the Standard Model of particle physics, general relativity, and ΛCDM cosmology — are empirically confirmed with extraordinary precision within their own domains, yet each of them leaves open questions it cannot close on its own. What is more, these open questions are beginning to interlock with one another in a way that suggests the problem is not the individual gaps in the theories, but a missing overarching framework.

The first crisis is a cosmological crisis. The Hubble constant — the fundamental parameter describing the expansion rate of the Universe — takes significantly different values when measured locally (via Type Ia supernovae, gravitational lenses, megamasers) and when derived from the cosmic microwave background observed by Planck. The discrepancy amounts to about 9% and, despite decades of systematic effort, does not go away. The standard ΛCDM model, based on dark energy in the form of the cosmological constant Λ and cold dark matter, works well, but it requires that 95% of the Universe's energy consist of two quantities whose physical nature we do not know. In addition, observations by the James Webb Space Telescope have revealed unexpectedly massive, mature galaxies and supermassive black holes already in existence 300–400 million years after the Big Bang — at times when standard models of structure formation struggle to accumulate sufficient mass.

The second crisis is a crisis of the foundation. Measurements of the mass of the Higgs boson (~125.1 GeV) and of the top quark (~172.7 GeV) place our Universe in the region of metastability of the Higgs potential: our quantum vacuum is, with high probability, not the state of the lowest possible energy, but merely a local minimum, separated from a deeper minimum by an energy barrier. At every point in space and time there exists a nonzero (though astronomically small) probability that quantum tunneling will initiate a phase transition — a bubble of new vacuum

expanding at the speed of light and rewriting the laws of physics. This is not a peripheral speculation: it is a direct consequence of the measured masses of the Standard Model particles, inseparably bound to the best-confirmed theory of the microworld.

The third crisis is a crisis of meaning. Quantum information theory, at the heart of quantum mechanics and black hole cosmology, says that information cannot be annihilated — it can only be transformed. Stephen Hawking devoted the last decades of his life to the black hole information paradox: if black holes evaporate thermally, the information contained in the matter they swallow should perish, which violates the unitarity of quantum mechanics. The contemporary consensus moves toward a resolution in which information is nonetheless preserved — either in subtle correlations of the Hawking radiation, or in the very geometry of the horizon (the holographic principle of 't Hooft and Susskind), or in a mechanism not yet identified. Regardless of the specific resolution, the fundamental principle — information is a conserved quantity — is becoming the third pillar of physics alongside energy and momentum. This gives rise to a question that standard physics does not take up but that cannot be ignored: if information is physically real and indestructible, what happens to the most highly organized information we know — conscious information — when its biological carrier dies?

The Cosmic Theory of Consciousness and Universe Inheritance is an attempt at an answer that joins these three crises within a single explanatory scheme. The central thesis reads: Universes constitute an evolving population in which each new cosmos inherits the physical parameters and the accumulated quantum information — including consciousness-information — from the previous generation. The mechanism of inheritance is the black hole singularity (an extension of Smolin's hypothesis) or the phase transition of the quantum vacuum from a false to a true minimum. Dark energy is, on this account, a manifestation of the "pressure" of daughter universes on the parent cosmos and of the accumulated consciousness-information encoded in the vacuum field. The Hubble tension is the signature of the growth of this pressure over time. The fine-tuning of the fundamental constants is the cumulative effect of many generations of cosmic selection.

The theory in its present shape arose in two layers. The first — the manifesto of May 2025 — sketched the philosophical framework and the basic architecture. The second — the Annex "the Transition Protocol" of March 2026 — introduced the physical mechanism in the form of a controlled or natural vacuum phase transition, the concept of consciousness-information as the building material of the new cosmos, the geometry of the wheel instead of the tree, and the two-path structure: the Controlled Phase Transition versus Natural Saturation. In April and May 2026, two independent scientific threads — the result of Slotine and Lohmiller on the classical reconstruction of the Schrödinger equation and the 4σ deviation in the LHCb experiment — provided both a mathematical bridge and an experimental suggestion of a gap in the Standard Model into which new physics could fit.

The present work is a synthesis of these layers. It does not claim the status of a formalized physical theory with a complete mathematical apparatus; it is, rather, a coherent hypothesis — a skeleton meant to be precise enough that it can be criticized, falsified, or developed. The author is an independent researcher, aware of the limitations this imposes. But as Lee Smolin wrote in "The Life of the Cosmos", even flawed speculative hypotheses have value if they organize questions in a way that leads to new observations and experiments. That is precisely what this synthesis is meant to be.

The text is divided into eight parts. Part I establishes the empirical and theoretical foundations — it presents the three crises of contemporary cosmology, the structure of the quantum vacuum, and the status of the metastability of the electroweak vacuum. Part II reconstructs the architecture of the theory — from the original metaphor of the tree to the more mature geometry of the wheel of universes, with black holes and the phase transition as parallel generative mechanisms. Part III develops the central thesis of consciousness as quantum information, based on the Penrose–Hameroff hypothesis, the work of Keppler, and Tononi's integrated information theory. Part IV describes the mechanism of the transition: the false vacuum, the geometry of the Higgs potential, the Slotine–Lohmiller mathematical bridge, and the concept of the dynamic Transition Point in the cosmic web of filaments. Part V presents the three paths — the Controlled Phase Transition, Natural Saturation, and Mitotic Saturation (introduced in version 5.0 as the

reproductive division of the universe through the saturation threshold, with the inheritance of learned information without the destruction of the parent) — together with their moral-cosmological implications. Part VI formulates the program of empirical tests, the seven groups of observational predictions (the seventh introduced in version 5.0 for testing reproductive signatures), and the criteria of falsification.
Part VII (introduced in version 3.0 and reformulated in version 4.0) contains the layer of mathematical operationalization: the reformulation of Keppler's hypothesis into the language of quantum electrodynamics, the consciousness-information functional $\mathcal{J}(t)$ within the frame of actualized potential, the concept of consciousness as the fraction of ZPF energy resonantly actualized by the biological network, the parameter η with its dual interpretation (the fraction of actualized potential and the degree of epigenetic modulation of the expression of the vacuum code — an extension from version 5.0), the phenomenological ansatz Λ_eff(t) for dark energy, and order-of-magnitude targets for observable consequences. Part VIII introduces a response to the consolation objection against the theory — in the form of three epistemic moves increasing its scientific resilience.
The work closes with a chapter relating CTCUI to other existing theoretical frameworks — multiverse cosmology, the holographic hypothesis, the vacuum landscape of string theory, panpsychism — and a short philosophical epilogue.

PART I — EMPIRICAL AND THEORETICAL FOUNDATIONS

1. Open problems of contemporary cosmology

1.1 The Hubble tension

The Hubble constant H_0 describes the present expansion rate of the Universe and is one of the most fundamental cosmological parameters. It is also one of the best measured. What has begun to emerge in recent years is therefore all the more unsettling: two independent classes of measurement systematically yield different values.
Local measurements, based on the cosmic distance ladder — Cepheids, Type Ia supernovae, gravitational lenses, and megamasers — give values around 73 km/s/Mpc with an uncertainty of less than 1 km/s/Mpc. Cosmological measurements, based on the power spectrum of the cosmic microwave background observed by Planck and derived under the assumption of the standard ΛCDM model, give values around 67 km/s/Mpc, likewise with very small uncertainty. The difference amounts to about 9% and reaches a statistical significance of the order of 5σ — which in physics corresponds to the threshold of "discovery." Dozens of independent analyses, checks for systematic errors, alternative calibrators of the distance ladder — none of this eliminates the tension.
The standard interpretation of the Hubble tension is that either one of the two sets of measurements carries an undiagnosed systematic error, or the ΛCDM model is incomplete — and new physics is required between the epoch of recombination and the present day, physics that alters the expansion rate in a way the standard model does not predict. Proposals range from "early dark energy" (a short-lived injection of dark energy around recombination) to modifications of gravity on large scales. None of these proposals has so far gained consensus.
CTCUI offers an alternative hypothesis: the discrepancy is the signature of the gradually growing influence of daughter universes on the geometry of our cosmos. If dark energy is not a cosmological constant but a dynamical pressure growing with the number and mass of daughter universes arising from black hole singularities, then we should expect its effective value measured locally (where the population of massive black holes in our vicinity is significant) to differ from the value derived from the times of recombination (when no black holes yet existed).

The prediction is qualitative but directed: the tension should persist and be systematic rather than statistical in character — which is exactly what we observe.

1.2 Dark energy and dark matter

The best current measurements of the energy composition of the Universe, based on observations of the cosmic microwave background, the large-scale structure of galaxies, and supernova data, yield the following picture: ordinary baryonic matter (atoms, nuclei, electrons) constitutes about 5% of the total energy density; dark matter — a substance interacting gravitationally but not entering into electromagnetic interactions — constitutes about 27%; and dark energy — a component with negative pressure, responsible for the accelerating expansion — constitutes about 68%. The two largest components of the Universe's energy are therefore physically unknown.

Dark matter has strong empirical support: galaxy rotation curves, gravitational lensing (especially in the Bullet Cluster), baryon acoustic oscillations, and the CMB spectrum all require its presence in very precise amounts. But decades of direct searches for dark matter particles — the XENON, LUX, PandaX, and SuperCDMS experiments, accelerator searches at the LHC — have so far produced no unambiguous signal. The most popular candidate — the WIMP, the weakly interacting massive particle — is being ever more strongly excluded.

Dark energy is more puzzling still. In the simplest interpretation it is simply the cosmological constant Λ, a term added to Einstein's equations describing the energy of "empty" space. But when one tries to compute its value from quantum field theory — by summing the zero-point energy of all vacuum modes — one obtains a value about 120 orders of magnitude larger than the observed one. This is the greatest discrepancy between theory and observation in the entire history of physics, known as the cosmological constant problem. Either all of that gigantic vacuum energy must magically cancel out, leaving a tiny residue that we see as dark energy, or dark energy is not simply vacuum energy — it is something else, for instance an evolving scalar field (quintessence) merely behaving like vacuum energy.

CTCUI proposes a third interpretation. Dark energy is a manifestation of two complementary phenomena: first, the pressure exerted on the geometry of our Universe by the daughter universes arising from black hole singularities; second, the coherent quantum information accumulated in the vacuum field by the forms of consciousness that have appeared in the course of cosmic evolution. Dark matter, within the same frame, may be understood partly as the gravitational effect of structures in dimensions to which we have no direct observational access (descendant universes developing in other "leaves" of the cosmic tree), and partly as conventional particulate dark matter in forms not yet identified. The theory does not claim to explain 100% of dark matter — it suggests, rather, that at least part of the gravitational anomalies may be a signature of the influence of daughter universes.

1.3 The information paradox

In 1974 Stephen Hawking showed that black holes should evaporate — emitting thermal radiation arising from quantum effects near the event horizon. The consequence is dramatic: if Hawking radiation is purely thermal, then a black hole that swallows a specific quantum state of matter and subsequently evaporates "forgets" everything about that state. All the information contained in the swallowed objects would be annihilated. This would violate one of the most fundamental principles of quantum mechanics — unitarity, that is, the conservation of information in the evolution of a quantum state.

The conflict between gravity and quantum mechanics at the point of information is so sharp that Hawking devoted a substantial part of his scientific life to it. In 2004 he publicly changed his mind relative to his earlier work, accepting that information is preserved after all — but the mechanism of that preservation remained, and remains, in dispute. The contemporary consensus, based on the holographic principle of 't Hooft and Susskind and on Maldacena's AdS/CFT correspondence, suggests that information is encoded on the surface of the event horizon (the holographic principle) and leaks out in subtle correlations of the Hawking radiation. Recent work on "islands"

in evaporating black holes (Penington, Almheiri, and others) has provided a concrete mechanism by which information can return to the environment in a unitarity-preserving manner.
CTCUI adopts an extended version of this resolution. The information preserved near the horizon is not merely passively encoded on the surface — it is actively transmitted to the newly forming daughter universe at the singularity. The black hole, on this account, is a two-sided information channel: the surface encoding is preserved with respect to our Universe (in accordance with the holographic principle), while the deep volumetric encoding becomes the "genetic code" of the descendant universe. This resolves the paradox without requiring a choice between unitarity and regularity of the geometry: information is preserved doubly — on the surface for the old cosmos and in the volume for the new one.

1.4 The fine-tuning of the fundamental constants

The standard list of about 26 independent fundamental constants of the Standard Model (together with the neutrino mass and mixing sector), supplemented by the gravitational constant and the cosmological constant Λ from general relativity, comprises the fermion masses, the coupling constants of the fundamental forces, the fine-structure constant α, the Higgs boson mass, the cosmological constant Λ, quark and neutrino mixing, and the CP-violation parameters. The discovery that surprises many physicists is that a great many of these constants must take values lying within narrow ranges for the Universe to be capable of forming stable matter and — ultimately — life.
A few examples. If the weak interaction were somewhat weaker, neutrons would not decay quickly after the Big Bang and hydrogen would not have formed. If the strong nuclear interaction were a few percent stronger, two protons would bind into a stable diproton and hydrogen would have burned up quickly, before galaxies could form. If the fine-structure constant α differed by a few percent, carbon would not have formed in stars in sufficient quantities (the Hoyle resonance in the carbon-12 nucleus is extraordinarily sensitive to α). If the cosmological constant Λ were larger by two to three orders of magnitude (the Weinberg bound, 1987), the expansion would have torn apart the condensations of matter before galaxies could form.
The standard answers to this fine-tuning are two. The first — the anthropic principle — holds that we observe life-friendly parameters simply because we are alive: in a multiverse with an enormous number of universes with randomly varying parameters, we must find ourselves in one of the few that permit life. The second — the string multiverse theory — proposes a vacuum landscape of about 10^{500} possible configurations of the geometry of the extra dimensions, each yielding different physical constants. Together, these two answers account for fine-tuning in a way that requires no purposiveness.
CTCUI proposes a supplement. The multiverse exists, but it is not random. It is the product of cosmic natural selection — universes that generate black holes (and, by extension, consciousness) reproduce; those that do not end their history without offspring. Over the course of many generations the fundamental constants converge toward values that optimize reproductive success. Fine-tuning is therefore neither a coincidence (classical anthropics) nor an infinite random search (the string landscape), but the cumulative effect of the evolutionary history of the multiverse.

1.5 The hard problem of consciousness

The hard problem of consciousness, formulated in its present form by David Chalmers in the 1990s, concerns the question of why physical processes in the brain are associated with subjective experience at all. We can describe in detail how neurons process information, how the brain generates behavior, how various brain states correlate with various reported experiences — but we have neither a theory nor even a consensus as to why there is "something it is like" to be a brain processing that information at all. What Chalmers calls "qualia" — the qualitative character of the experience of pain, of redness, of taste — remains fundamentally mysterious.
The standard materialist answers propose that consciousness is an emergent effect of complex informational organization (functionalism), an illusion (eliminativism), or simply a non-problem

amenable to direct resolution by neuroscience (mysterianism). Alternative answers range from panpsychism (consciousness is a fundamental feature of the universe, present at every level of the organization of matter) to dualism (consciousness is a separate substance or property irreducible to physics).

In recent decades, Giulio Tononi's integrated information theory (IIT) has emerged, proposing a mathematical measure of consciousness — Φ (phi), a number encoding the degree of integrated information in a system. A system is conscious if it has nonzero Φ, and the amount of consciousness is proportional to the magnitude of Φ. IIT leads logically to panpsychism — every system with nonzero Φ has some degree of consciousness — but it also precisely distinguishes human consciousness from, for example, the consciousness of a classical computer (which has Φ = 0 or very small Φ, despite great computing power).

Independently, the Penrose–Hameroff hypothesis (Orchestrated Objective Reduction, Orch-OR) suggests that consciousness arises from quantum processes in neuronal microtubules, where the reduction of a quantum superposition yields a moment of subjective experience. The hypothesis was long criticized as biologically implausible on account of decoherence, but recent experiments (Bandyopadhyay and others) have demonstrated the existence of quantum oscillations in microtubules at body temperature.

The most recent proposal, on which CTCUI draws heavily, was formulated by Joachim Keppler (2025): the brain does not generate consciousness in isolation but resonates with the quantum zero-point field (ZPF), with glutamatergic neurotransmitters in the cerebral cortex resonating with specific ZPF modes. Consciousness is thus a field phenomenon — the brain is a "receiver" or resonator, not a generator. This opens the way to interpreting consciousness as a fundamentally quantum property of the vacuum field — a property that exists in the field independently of particular biological carriers, though it needs such carriers in order to manifest locally.

CTCUI adopts a synthesis: consciousness is quantum information encoded in the zero-point field; local living organisms are its resonators; every act of conscious experience generates unique quantum correlations in the field, which do not perish upon the death of the carrier but integrate into the global informational structure of the Universe. Tononi's integrated information (Φ) provides the measure, the Penrose–Hameroff hypothesis provides the microphysical mechanism, and Keppler's proposal provides the bridge between the brain and the vacuum field.

2. The quantum vacuum: the foundation

2.1 What really fluctuates

The notion of the "vacuum" has an entirely different meaning in quantum physics than in everyday intuition. In quantum mechanics the fundamental reality is not particles but fields — a continuum of values of some physical quantity extending through all of space. The electron is an excitation of the electron field, the photon of the electromagnetic field, the quark of the quark field; every known particle corresponds to some quantum field. "Vacuum" denotes the state in which none of these fields has any excitations — no particles — and it is the state of the lowest possible energy for a given configuration of fields.

But "lowest possible" does not mean "zero." Every quantum field, in accordance with Heisenberg's uncertainty principle, must have a nonzero ground-state energy — a zero-point energy, which follows from the fact that the field's amplitude and its rate of change cannot both be exactly zero at once. This is the "trembling" of the field, which persists even in "empty" space and cannot be eliminated.

In the particle picture, this zero-point trembling of the fields is sometimes described as the constant appearance and disappearance of virtual particle–antiparticle pairs — momentary excitations permitted by the uncertainty principle as long as their lifetimes are sufficiently short. Electron–positron pairs, quark–antiquark pairs, virtual photons — all of these excitations constantly arise and vanish in the vacuum. The language of virtual particles is, however, a

computational metaphor used in perturbation theory and Feynman diagrams; the underlying reality is the fluctuating fields and their energy, not literal particles flying through space.
Experimental confirmation of the existence of these fluctuations is provided by the Casimir effect. Two parallel metal plates placed in vacuum at a separation of the order of nanometers attract each other with a force that cannot be explained by classical electrostatics — the plates are electrically neutral. The quantum explanation is as follows: between the plates only those modes of the electromagnetic field can exist that "fit" into the gap as standing waves, which restricts the number of allowed modes and their total zero-point energy. Outside the plates no such restriction applies — the vacuum fluctuations have greater "freedom" — and the resulting difference in mode pressure yields a compressive force. Measurements have shown agreement with the theoretical predictions of QED to better than 1%. The vacuum has a real, measurable structure.

2.2 Vacuum energy and the cosmological constant problem

The total zero-point energy of all modes of all quantum fields is, in a naive calculation, infinite — summing over infinitely many modes of increasing frequency gives a divergent sum. The standard procedure here is to cut the sum off at the energy scale above which our theories cease to apply — most commonly the Planck scale, about 10^{19} GeV. After the cutoff we obtain a vacuum energy density of the order of 10^{113} J/m^3.
The observed density of dark energy (assuming it is a cosmological constant) is about 10^{-9} J/m^3. The ratio is $\sim 10^{120}$ — one hundred and twenty orders of magnitude. This is the greatest discrepancy between theory and observation in the history of physics.
Two schools of reaction are possible. The first accepts that the theoretical vacuum energy really is that large, but a gigantic cancellation between positive and negative contributions (from bosonic and fermionic fields, respectively) leaves only a small residue. The mechanism of that cancellation is not known — supersymmetry could provide it, but we observe no supersymmetric partners at the LHC. The second school questions the very interpretation of dark energy as vacuum energy — it proposes a scalar field (quintessence) or a modification of gravity.
CTCUI proposes a third way. Dark energy is indeed connected with the vacuum field, but its observed value is not the final "residual" vacuum energy — it is a dynamical function, dependent on (a) the pressure of daughter universes on the geometry of our cosmos and (b) the accumulated coherent consciousness-information encoded in the vacuum field. On this account the enormous theoretical vacuum energy is real, but its effective manifestation as dark energy is modulated by two evolutionary processes on the cosmological scale. Prediction: dark energy should evolve in cosmological time, and its rate of evolution should correlate with the density of structures generating coherent information — galaxies, clusters, populations of conscious beings.

2.3 Why the vacuum is more fundamental than matter

The standard materialist picture of physics treats particles as the basic "building blocks" of reality, and the vacuum as "nothing" — the space in which the blocks are located. Modern quantum field theory inverts this picture: the fields are fundamental, their excitations secondary. Particles are local ripples of the field; the vacuum is the state of the field without excitations, but with its full structure and energy.
This inversion has deep implications. If fields are fundamental, and the vacuum is the field's lowest state — nonzero, fluctuating, energetically active — then the vacuum is what fills the Universe "from the lining outward." We are not objects in a void; we are local configurations of fields against the rich background of a vacuum that constantly trembles, fluctuates, and generates energy.
CTCUI draws from this a central consequence: since the vacuum is the foundation, and consciousness is a physically real phenomenon, consciousness must in some way be grounded in the very structure of the vacuum. Not in neurons as such — neurons are too complex, too high-level as objects, to serve as the foundation of something that appears qualitatively basic, as experience does. Consciousness is a property of the vacuum field; the brain is its resonator.

3. Electroweak stability and the metastability of the vacuum

3.1 The landscape of the Higgs potential

The Higgs field, whose excitation is the Higgs boson discovered at the LHC in 2012, is a central component of the Standard Model. Its nonzero vacuum expectation value (vev, about 246 GeV) gives mass to the elementary particles through the Brout–Englert–Higgs mechanism. The Higgs boson, with a measured mass of about 125.1 GeV, is the visible excitation of this field.
The landscape of the Higgs potential — the graph of the potential energy of the Higgs field as a function of its value — has, in the simplest approximation, the shape of a "bottle bottom" (the Mexican hat potential) with a minimum at the vev. But when quantum corrections are taken into account — in particular loops involving massive particles, dominated by the top quark — the potential is significantly modified at very large values of the Higgs field. For the Higgs and top quark masses measured in experiment, the potential becomes unstable — at very high energy scales a second minimum appears, much deeper than the present one, separated from it by an energy barrier.
Theoretical calculations, presented most fully in the works of the groups of Markkanen, Rajantie, and Stopyra, indicate that our Universe is with high probability located in the metastable region: our vacuum is not the absolute minimum of the Higgs potential, but a false minimum, separated from the deeper, true minimum by a barrier that can be overcome by quantum tunneling.

3.2 Quantum tunneling and the lifetime of the vacuum

Quantum tunneling is a purely quantum phenomenon with no classical counterpart: a system can "jump" across an energy barrier it lacks the energy to overcome classically, with a probability depending on the width and height of the barrier. In the context of the Higgs vacuum, the "jump" means the spontaneous appearance, somewhere in space, of a small bubble of new, true vacuum — a bubble inside which the Higgs field has the value corresponding to the true minimum, and in consequence the laws of physics are different.
Calculations of the lifetime of our false vacuum are uncertain, but they yield values of the order of 10^{100} years or more — astronomically longer than the present age of the Universe ($\sim 1.4 \times 10^{10}$ years). In practice this means that spontaneous tunneling is not a source of immediate danger. But "nonzero" is mathematically different from "never." In infinite time, it is certain.
Moreover, the lifetime is a function sensitive to the values of the fundamental constants — in particular to the masses of the Higgs and the top quark. Precise measurements of these masses are among the main goals of the current LHC program. The present values place us in the metastable region, close to the boundary between metastability and stability, but not close enough to settle definitively which region we are in.

3.3 The transition bubble: physical consequences

If and when spontaneous tunneling occurs at some point in the Universe, a local bubble of true vacuum will begin to expand. The rate of expansion is limited only by the speed of light; the bubble wall — the region in which the Higgs field changes its value from the false to the true minimum — moves outward with $v \to c$. In the thin-wall approximation (for a bubble much larger than the wall thickness), the wall reaches the speed of light within the first moments of expansion.
Inside the bubble, new laws of physics hold. The coupling constants, the particle masses, the structure of the fundamental forces — all of these take values corresponding to the true minimum, not to our present false minimum. In practice this means that atoms as we know them do not exist inside the bubble — electrons have a different mass or are not stable at all; atomic nuclei disintegrate; chemical bonds vanish. Objects built of the old matter are instantly destroyed as the bubble wall passes through them.
From the perspective of an observer in a region the bubble has not yet reached, there is no warning whatsoever. An electromagnetic signal from the bubble does not outrun the bubble itself — it travels at the same speed. A telescope would not see the approaching wall, because the light

from the wall arrives exactly when the wall does. "The cleanest death in the Universe" — Ethan Siegel's description — is physically exact: no time for perception, no time for fear, no time for anything.

But — and this is the key point of CTCUI — what is destruction from the perspective of the old Universe is birth from the perspective of the new one. The energy released in the phase transition (the energy difference between the false and the true minimum, multiplied by the volume converted) is gigantic. In the Coleman–De Luccia model this energy is redistributed: part supplies the kinetic energy of the expanding wall, part generates particles and radiation inside the bubble — creating a hot, expansive phase in which new particles, antiparticles, and radiation dominate the dynamics, much as in the hot Big Bang.

The bubble of new vacuum is thus a new Universe at the stage of primordial cosmology. From the perspective of the old Universe it is annihilation; from the perspective of the new one — birth. This dual reading of the same event is the foundation for the rest of the theory.

3.4 The location of ignition

In the standard description of quantum tunneling, the probability of a bubble's "ignition" is uniform in space — at every place in the Universe there exists the same baseline chance of a spontaneous transition. It is a fundamentally random and nonlocal phenomenon, unrelated to the presence of an observer or the presence of matter.

Despite this uniformity, theory points to several conditions or places where the probability of ignition may be locally modified. First, extreme energy densities — the vicinity of black hole singularities, where quantum fields are subjected to extreme conditions, may locally modify the shape of the Higgs potential. Some models suggest that black holes could act as catalysts of the transition. Second, radiation of very high energy — particle collisions at energies comparable to the Planck energy could locally excite the Higgs field over the potential barrier, but such energies are unattainable for current or planned accelerators (the LHC operates at $\sim 10^4$ GeV; the Planck energy is $\sim 10^{19}$ GeV). Third, the very early Universe — the temperature just after the Big Bang was so high that the Higgs field could have oscillated near the top of the potential barrier; the fact that no transition occurred is one of the arguments for the stability of our vacuum. Fourth, regions of exceptionally low matter density (the great cosmic voids) — some models suggest that where matter and energy are minimally present, the Higgs field has the fewest stabilizing "perturbations," though this line of argument is speculative.

CTCUI adds a fifth factor, which does not appear in standard physics: the coherent structure of quantum consciousness-information encoded in the vacuum field. The argument runs as follows: since the Casimir effect shows that the geometry of field modes locally changes the vacuum energy, a sufficiently coherent, organized fluctuation of the field on the cosmic scale should be capable of locally modifying the shape of the Higgs potential. This will be developed in detail in Part IV.

3.5 Empirical status: the LHCb anomaly

In April 2026 the LHCb experiment published a result that may prove to be one of the more important in physics in recent decades. Measurements of the angles, energies, and rates of the so-called "electroweak penguin decay" of B mesons (a process in which a B meson transforms into four particles: a kaon, a pion, and two muons) showed a deviation from the Standard Model predictions at the level of 4σ — corresponding to a 1 in 16,000 chance that the deviation is merely a statistical fluctuation.

The result does not yet reach the conventional "discovery" threshold (5σ), but it is significant for two reasons. First, it is independently corroborated — the CMS experiment published a consistent measurement earlier in 2025, though with lower precision. Second, it concerns an extraordinarily rare process (the decay occurs once per million B mesons) and is for that reason especially sensitive to the influence of unknown particles or forces. The most widely discussed explanation is the so-called "leptoquarks" — theoretical particles linking leptons and quarks — but other scenarios remain open as well.

For CTCUI, the significance of this result is as follows. The Standard Model, for all its successes, is incomplete — it does not account for gravity, dark matter, or dark energy. Every confirmed "crack" in its structure opens space for new physics. If leptoquarks or other hitherto unknown particles really do exist at energy scales accessible to the LHC, then our knowledge of the structure of quantum fields — including the exact geometry of the Higgs potential, the stability of the vacuum, and the nature of the interactions between fields — requires revision. The LHCb anomaly is not by itself evidence for any specific aspect of CTCUI, but it is an empirical reminder that the picture we have today is almost certainly not final. CTCUI flourishes in that crack — in the space the Standard Model leaves open.

PART II — ARCHITECTURE OF THE THEORY

4. Universes as a population: from the tree to the wheel

4.1 The original metaphor: the tree of universes

In the original formulation of the Cosmic Theory of Consciousness and Universe Inheritance (CTCUI, May 2025), universes were metaphorized as leaves of a cosmic tree: our Universe — a single leaf, black holes — seeds generating new branches and new leaves, each descendant universe with slightly modified physical parameters relative to its parent. The tree metaphor was intuitively capacious, because it captured three essential aspects: historical continuity (the descendant comes from the parent), differentiation (each leaf is unique, though similar), and expansion (the tree grows over time).

The tree, however, has a certain feature that proved problematic in the course of developing the theory: it implies a beginning. If universes are leaves on a tree, the tree must have a trunk, and the trunk must have a root — a first Universe that came from no parent. This raises the same metaphysical problems that afflict the standard Big Bang model: where did this first Universe come from, why does it have these parameters rather than others, does its existence require some metaphysical assumption beyond the theory itself.

4.2 A more mature geometry: the wheel of universes

The annex "the Transition Protocol" (March 2026) proposed moving from the metaphor of the tree to the metaphor of the wheel. In a wheel there is no beginning and no end — each Universe gives birth to the next, which gives birth to the next, in a cycle whose very structure requires no metaphysical root. The question "what came first" presupposes linear time; the wheel as a geometry corresponds to a cyclic, nonlinear conception of the generation of the cosmos.

The wheel does not mean a repetition of the same universes. Each new cosmos in the cycle is new — it has slightly different parameters, a greater density of accumulated information, more refined conditions for the development of complex structures. The geometry of the wheel combines continuity (each Universe has a parent) with evolution (each new one is different and potentially better). This is the geometry corresponding to natural selection on a cosmological scale: a cycle of reproduction with inheritance and variation.

Concretely: one can imagine the structure of the multiverse as a multilayered spiral in which each parent universe generates many descendants through various black holes (and potentially through vacuum phase transitions), and the descendants in turn generate their own descendants. The multiverse is a directed graph in which the nodes are universes and the edges are generative relations. This graph has no single "root"; each Universe is a node that is both a source and a target of many edges. The topology of the whole is cyclic at a sufficiently large scale — each branch ultimately interlocks with other branches through shared informational foundations.

4.3 Evolution of parameters

Each new Universe inherits physical parameters — particle masses, coupling constants, the cosmological constant, the geometry of extra dimensions (if they exist) — from the parent universe, but with small modifications. The modifications are analogous to genetic mutations: small, random shifts in parameter values, induced by quantum fluctuations at the moment of the generative transition (a black hole or a vacuum phase transition).
Those universes that inherit parameters favorable to the creation of black holes (and, by extension, of consciousness) reproduce — they generate new offspring. Those that inherit unfavorable parameters — expansion too fast for structures to form, a cosmological constant too large, gravity too weak to form stars — end their history with no offspring, or with minimal offspring. Over the course of many generations, the population of the multiverse becomes dominated by universes with parameters that optimize reproductive success.
This is the mechanism of cosmological natural selection, proposed in its original form by Lee Smolin in "The Life of the Cosmos" (1997). CTCUI adopts it as a foundation and extends it in one essential respect: in addition to physical parameters, quantum consciousness information is also inherited, which adds a new dimension to the selection process.

5. Black holes as generative transitions

5.1 Smolin's hypothesis and its extension

Lee Smolin proposed in 1992 that the singularities of black holes — points in spacetime where curvature grows to infinity and classical general relativity ceases to apply — are in reality the birthplaces of new universes. Instead of a true singularity, where the mathematics "blows up," a quantum process generates a new, separate spacetime with its own Big Bang, connected to our Universe through the "umbilical" of the singularity.
The physical mechanism of this hypothesis is not fully worked out — it would require a theory of quantum gravity, which to this day we do not possess in an experimentally verified version. But the general structure is consistent with several advanced proposals. Loop Quantum Gravity (LQG) suggests that classical singularities are in reality "quantum bounces" — at very high densities gravity becomes effectively repulsive and prevents a true singularity; the energy is "bounced" into a new geometry. "Bouncing cosmology" models develop this idea, treating the Big Bang as a bounce from a previous cycle of contraction. CTCUI brings these threads together into a single hypothesis: every black hole singularity is a "bounce" generating a new daughter universe.

5.2 What passes through the singularity

The critical question is what exactly passes from the parent universe to the daughter universe through the black hole singularity. The naive answer — "all the matter that fell into the black hole materializes in the new Universe" — is too crude, because it ignores the subtlety of the quantum state of matter and information.
The more precise answer that CTCUI proposes is as follows. Matter as local excitations of quantum fields does not pass through in any simple way — it undergoes a transformation into a primordial, high-energy phase, similar to the plasma immediately after the Big Bang. But the quantum information contained in the absorbed state — all the correlations, entanglements, parameters characterizing quantum microstructures — is preserved, in accordance with the principle of information conservation. This information becomes the material defining the initial conditions of the new Universe.
In particular, if the theory of holographic information is correct, the information encoded on the surface of the event horizon (and thus, in a certain sense, all the details of everything that ever crossed that horizon) becomes the skeleton for the geometry of the new Universe. The physical parameters of the new cosmos — particle masses, coupling constants, the structure of the fundamental forces — are a function of the accumulated information in the "parental" black hole. This explains why descendant universes have parameters close to the parent's, but not identical:

they are based on the same informational foundation, but with modifications arising from the dynamics of the singularity itself.

5.3 JWST observations and the black hole population in the early Universe

Observations by the James Webb Space Telescope have revealed a surprising phenomenon: supermassive black holes already existed 300–400 million years after the Big Bang, at times when standard models of black hole formation (through accretion onto regular stellar "seeds") struggle to accumulate sufficient mass. Some of these objects have masses on the order of 10^8–10^9 solar masses, which is difficult to explain at a time when galaxies were only just forming.
Standard explanatory proposals include: the direct collapse of massive gas clouds (so-called "direct collapse black holes"), primordial black holes formed in the very early Universe from density fluctuations, or accelerated accretion rates in the first hundreds of millions of years. Each of these proposals is still under debate.
CTCUI adds a hypothesis: some of the early supermassive black holes may be an "inheritance" from the previous cosmological cycle — informational structures that passed through from the parent universe and constituted themselves in ours as initial density asymmetries around which matter condensed. This hypothesis is speculative, but qualitatively testable: if the early supermassive black holes really are an inheritance, their spatial distribution and mass statistics should differ from the population forming de novo in our Universe. In particular, they should be relatively uniformly distributed (as a global informational structure), rather than concentrated in the centers of primordial matter overdensities.

6. Cosmological natural selection and its informational extension

6.1 Classical cosmological natural selection (Smolin)

Smolin's hypothesis has an elegant logical structure. Assume that (a) black holes generate new universes, (b) descendants inherit the parent's parameters with small mutations, (c) some combinations of parameters favor the creation of black holes, others do not. The consequence is immediate: after many generations, the population of the multiverse is dominated by universes with parameters that optimize black hole production.
This leads to a testable prediction: the parameters of our Universe should be "close to optimal" for black hole production. Smolin argued that this is indeed the case: small changes in most of the fundamental constants would reduce the number of black holes formed. This argumentation was and is debated — some physicists have raised counterexamples — but it has not been unambiguously refuted.
The known counterexamples to pure black-hole selection — combinations of parameters that would increase black hole production yet are not realized in our Universe — CTCUI takes as data, rather than passing over in silence. In the informational extension (chapter 6.2) the fitness function is not purely black-hole-based: selection rewards universes that generate both black holes and consciousness, and these two goals are not maximized by the same parameters. Counterexamples to Smolin are therefore expected within CTCUI: our cosmos should look black-hole suboptimal precisely to the extent that it "pays" for its capacity for complexity and consciousness.
An important test of Smolin's hypothesis is the existence of supersymmetry. If supersymmetry exists at directly accessible energy scales, Smolin's model is violated, because supersymmetric universes would produce fewer black holes than ours. The LHC has so far found no supersymmetric partners — which Smolin interprets as support for his hypothesis.

6.2 The informational extension in CTCUI

CTCUI adopts Smolin's hypothesis as a mechanistic foundation and extends it in one essential dimension: in addition to physical parameters, the quantum information of the parent universe is also inherited, including information associated with consciousness.
The mechanism is as follows. Consciousness, as a fundamental excitation of the vacuum field (see Part III), generates coherent quantum correlations that are encoded in the global informational structure of the Universe. Over the lifetime of the parent universe these correlations accumulate — every act of conscious experience adds its unique signature. When matter and information pass through black holes (or through a vacuum phase transition), this accumulated information becomes an integral part of the code defining the new Universe.
The consequence is that selection in CTCUI is not merely selection for black hole production — it is also selection for the capacity to generate consciousness. Universes that generate black holes but do not generate consciousness pass on physical parameters, but pass on an empty informational field. Universes that generate both black holes and consciousness pass on richer information — their descendants have better initial conditions for the development of complexity and of further generations of consciousness.
Over many generations, universes become ever more "tuned" to generating consciousness — not because this favors black hole production (which is the engine of Smolin's classical selection), but because consciousness in itself enriches the informational foundation of inheritance. Each successive Universe has more to pass on to the next, because it carries the history of consciousness information from all previous generations.

6.3 The biological analogy

The analogy with biological evolution is instructive, but requires caution — it should not be stretched beyond the places where it is substantively similar.

Biological evolution	**Cosmological evolution in CTCUI**
Organisms inherit DNA	Universes inherit physical parameters and quantum information
Genetic mutations	Quantum fluctuations of parameters at the moment of the generative transition
Natural selection favors organisms adapted to their environment	Cosmological selection favors universes capable of generating black holes and consciousness
Biological complexity grows in the course of evolution	Cosmic complexity grows across generations of universes
Evolutionary information is written in genes	Cosmic information is written in the fundamental constants and the vacuum field
Organisms develop increasingly complex structures	Universes develop ever better conditions for the emergence of consciousness
Selection acts through differential reproductive success	Cosmic selection acts through the differential production of daughter universes

A key caveat: this analogy does not imply that cosmological evolution is "purposeful" in a sense in which biological evolution is not — in both cases the mechanism is blind in the sense that it presupposes no "designer." Nonetheless, in both cases the structure of the process leads to a growth of complexity, and in the case of CTCUI — a growth in the capacity to generate consciousness. One may call this growing tendency a "direction," without presupposing causal intention.

6.4 The second mechanism: a controlled phase transition

Smolin's classical cosmological natural selection and its informational extension in CTCUI operate on an automatic, "natural" generative mechanism — black holes arise in the course of ordinary stellar and galactic evolution, and their singularities generate daughter universes without the involvement of any intention. The annex "the Transition Protocol" introduced, however, a second generative mechanism, which is not "natural" in the same sense: a controlled phase transition of the vacuum from the false to the true minimum.

The physical mechanism is described in detail in Part IV. Here it suffices to note the structural difference: black holes are generated "involuntarily," by stellar processes. A vacuum phase transition can be generated "deliberately," by a sufficiently advanced civilization commanding sufficient energy and knowledge of the topology of the vacuum potential field. From the perspective of cosmological selection, this adds a new dimension: in some generation of universes, evolution ceases to be a blind process and becomes an intentional one.
The biological analogy is telling: biological evolution on Earth was for billions of years a process of blind natural selection, but in recent decades it has begun to give way to conscious genetic engineering. From an evolutionary perspective, a species that has mastered CRISPR is a qualitatively new type of agent — it can modify the hereditary information of its offspring intentionally. An analogous transformation on the cosmic scale would mean that a certain civilization had passed from being a product of cosmic evolution to being its co-designer.
CTCUI leaves open the question of whether such a transformation has ever taken place (or ever will take place) in our Universe, or of which generation it would be the product. The hypothesis that it has taken place is the foundation for the Controlled Transition scenario (Part V), but it is not the only scenario. The alternative is Natural Saturation — a path in which universes stabilize through the accumulation of consciousness without the need for a controlled transition. Both paths are equally legitimate consequences of the theory.

PART III — CONSCIOUSNESS AS QUANTUM INFORMATION

7. Consciousness as a field excitation

7.1 A critique of computational materialism

The picture of consciousness dominant in cognitive neuroscience — computational functionalism — holds that consciousness is an emergent property of sufficiently complex information-processing systems. In this view the brain is a biological computer, consciousness is an algorithmic property of its operation, and in theory any system with the appropriate computational structure (including a sufficiently advanced digital computer) should be conscious.
Computational functionalism, however, is deeply challenged both empirically and conceptually. Empirically: decades of neuroscientific research have revealed no specific algorithm or computational structure that could be unambiguously identified as the "correlate of consciousness." The correlates are systemic, distributed, and far more complex than any simple computational model. Conceptually: Searle's "Chinese room" argument, the "qualia inversion" thought experiment, and many others show that identical algorithms can plausibly occur without consciousness or with different consciousness — which undermines the claim that algorithmic structure is a sufficient condition for experience.
CTCUI rejects computational functionalism as a foundation and adopts a panpsychist position within a quantum-field-based interpretation: consciousness is a fundamental property of the vacuum field, not an emergent property of computational organization. The brain is a specialized resonator that localizes and concentrates this global property, but does not generate it.

7.2 The Penrose–Hameroff hypothesis: quantum reduction in microtubules

Roger Penrose and Stuart Hameroff proposed in the 1990s Orchestrated Objective Reduction (Orch-OR) — the hypothesis that consciousness arises from quantum processes in the microtubules of neurons. Microtubules, cylindrical protein structures found in the cytoskeleton of every cell (including the neuron), can — according to this hypothesis — maintain quantum superpositions of the electrical polarization states of tubulins for times long enough that the

spontaneous reduction of the superposition (objective reduction, OR, in Penrose's sense) generates a moment of subjective experience.
The hypothesis was long criticized for two main reasons. First, quantum decoherence at body temperature should be immediate — the superpositions that Orch-OR requires should be destroyed within microseconds by interaction with the environment. Second, Penrose's mechanism of "objective reduction," which links the reduction of the superposition to quantum gravitational effects, is speculative and untested.
In recent years, however, experimental results have appeared that weaken both criticisms. Anirban Bandyopadhyay and his collaborators have shown that microtubules can support quantum resonant oscillations across many frequency scales and under biological conditions, which suggests that decoherence is less destructive than previously thought. Experiments with xenon as an anesthetic (Hameroff and others) suggest that conscious states are specifically coupled to quantum processes in microtubules — anesthesia acts by disrupting these processes.
Independent confirmation of quantum processes in biology comes from other areas: quantum coherence in photosynthesis (the FMO complex), quantum magnetoreception in birds (based on entanglement in cryptochromes), quantum tunneling in enzymes. Quantum biology is no longer merely a hypothesis — it is an established field of research in which quantum effects are engaged in biological processes at scales far larger than previously thought.

7.3 Keppler's hypothesis: brain resonance with the zero-point field

Joachim Keppler, in works from 2025, proposed a model that CTCUI regards as the most fully developed candidate for a bridge between the brain and the vacuum field. The central thesis is: the brain does not generate consciousness in isolation, but resonates with the quantum zero-point field (ZPF) — the same field whose fluctuations generate the Casimir effect.
The mechanism is as follows. Glutamatergic neurotransmitters in the cerebral cortex resonate with specific modes of the ZPF, leading to a state of self-organized criticality — a state in which brain activity balances on the edge between chaos and order, optimizing the capacity for information processing. In this critical state the brain acts as a resonator of certain ZPF modes, "selecting" them from the rich structure of vacuum fluctuations and locally amplifying them.
The conceptual consequence is fundamental. Consciousness is not a product of neuronal computation; it is a property of the field that neurons localize and concentrate. The brain is like a tuning fork, which does not generate sound waves but selectively resonates with specific frequencies in the surrounding air. The "air," in the case of consciousness, is the zero-point field — the omnipresent, fundamental structure of the vacuum.
This interpretation resolves the hard problem of consciousness in a way that requires neither materialist eliminativism (consciousness is not an illusion) nor metaphysical dualism (consciousness is not a separate substance). Consciousness is a physically real property of the field; brains are specific configurations of matter that resonate with this property; subjective experience is the result of that resonance. The hard problem does not disappear entirely — the question remains why the vacuum field has the property of consciousness — but it shifts to a level at which it can be addressed as a fundamental feature of physical reality, rather than as a mysterious emergent effect of brain complexity.

8. Integrated information as a measure

8.1 Tononi's IIT

Giulio Tononi's Integrated Information Theory (IIT) provides a mathematical measure of consciousness — integrated information Φ (phi). Φ is a function of an informational system: it measures how much information the system as a whole carries over and above the information carried by the sum of its parts. The higher the Φ, the higher the "quantity" of consciousness.
The consequences of IIT are far-reaching. First, it leads to a version of panpsychism — every system with nonzero Φ has some degree of consciousness. An atom, a photon, a drop of water —

all have an infinitesimally small, but nonzero Φ. Only systems with sufficiently large Φ — brains, ecosystems, potentially galaxies — have "a lot" of consciousness in the human sense. Second, IIT precisely distinguishes consciousness from computational power: a digital computer of enormous computational power may have Φ close to zero (because the information in it is weakly integrated between modules), while a brain of much smaller "computational" power has a very high Φ (because the information is densely integrated in the cortex).
CTCUI adopts IIT as the measure of the "quantity" of consciousness and combines it with Keppler's hypothesis of resonance with the vacuum field. The synthetic hypothesis: the brain's resonance with specific ZPF modes generates particular patterns of integrated information; Φ measures the richness and coherence of these patterns. A system of high Φ is both a resonator of a complex ZPF structure and a "record" of that structure in its local organization.

8.2 Φ as a contribution to the global field

CTCUI goes further than classical IIT and proposes that Φ generated locally (by brains, ecosystems, civilizations) is physically accumulated in the global quantum field. Every act of conscious experience generates coherent quantum correlations in the zero-point field that are not confined to the location of the generator. These correlations propagate — in a manner analogous to quantum entanglement, but on a larger scale — and accumulate in the global informational structure.
The key implication is that the quantity of consciousness in a given Universe is not simply the sum of the Φ of all individual brains; it is the integrated Φ of the entire field, which grows faster than linearly with the number of generators, provided those generators are sufficiently diverse. A homogeneous civilization — everyone generating similar patterns of consciousness — contributes less to the global Φ than a diverse civilization in which different forms of consciousness complement one another.
This is the physical reason why the diversity of conscious forms of life has cosmological, and not merely ethical, significance. Every new form of consciousness — based on a different chemical substrate, in a different planetary environment, with a different cognitive structure — contributes unique patterns to the global field. These patterns are not interchangeable — one cannot "manufacture" the pattern of one form of consciousness by multiplying another. The informational richness of the field is a function of the diversity of its sources.

9. Consciousness preserved: the inheritance of information

9.1 The principle of quantum information conservation

Quantum mechanics, in its standard formulation, is a unitary theory: the evolution of a quantum state in time conserves information. The evolution operator is unitary, which means that the full information about the initial state is always in principle recoverable from the final state (given knowledge of the operator). This principle of information conservation is one of the foundations of quantum mechanics, and it was what Hawking suspected black holes of violating.
The consequence of the principle is as follows: information, in the quantum sense, cannot be annihilated. It can be transformed, dispersed, entangled with the environment to the point of becoming practically unrecoverable — but fundamentally it remains. Even if we are never able to reconstruct it, it is there, somewhere in the full quantum state of the Universe.
CTCUI adopts this principle and draws out its consequence for consciousness: if consciousness is quantum information — coherent correlations in the zero-point field — then it cannot be annihilated by the death of its biological carrier. The brain is a resonator; when the resonator ceases to function (in the sense of biological death), the locally generated consciousness ceases to be localized, but the information it generated in the field does not vanish. It integrates into the global informational structure, where it remains as a permanent record of modified quantum correlations.

9.2 Nonlocality and entanglement at macroscopic scales

Classical quantum mechanics has shown that quantum entanglement — a situation in which two particles are correlated in a way that cannot be explained by any local model — is a fundamental property of physical reality. Experiments violating Bell inequalities, from the first attempts by Aspect and others to the most recent rigorous tests excluding local realism, have confirmed entanglement with unprecedented precision.

Entanglement, however, is nonlocal only in the sense of statistical correlations — it does not allow information to be transmitted faster than light, and it generates no observable "effects" of one particle on the other. Even so, it fundamentally indicates that physical reality has a deeper structure than a simple superposition of local events: two distant systems can be part of a single quantum state.

CTCUI assumes that nonlocal correlations of this kind also exist on a larger scale — between the brains of different organisms, between the ecosystems of different planets, between forms of consciousness in different historical periods. These macroscopic entanglements are not observable in current experiments, but their existence is not ruled out by known physics if we treat consciousness as a field effect. The zero-point field extends through all of space; its fluctuations are nonlocally correlated; consciousness, as a coherent excitation of this field, inherits these nonlocal correlations.

9.3 Accumulation of consciousness information in the vacuum field

Combining three elements — the conservation of quantum information, consciousness as a field excitation, macroscopic entanglement — CTCUI formulates its central hypothesis of the inheritance of consciousness information:

Every biological consciousness generates, over the course of its life, a set of coherent quantum correlations in the zero-point field. This set is unique to each organism — it is the "quantum signature" of its experiences, decisions, emotions. After the organism's biological death the local manifestation of consciousness ceases, but the correlations it generated remain in the field, where they integrate into the global informational structure. Over evolutionary time — billions of years of the existence of conscious forms of life — the global informational structure becomes ever richer.

This growing richness has no visible consequence in classical cosmology — the information is "silent," it generates no observable effects. But according to the conception developed in Part IV, when the density of integrated information in the field reaches a certain critical threshold, its influence on the geometry of the vacuum potential field can become nontrivial. Consciousness begins to affect the very structure of the vacuum that carries it. This is the moment at which the physics of the universe ceases to be neutral with respect to the experience it contains.

9.4 The Saturation Threshold: cosmic consciousness

The hypothesis introduced by the annex "the Transition Protocol" holds that there is a logical threshold — named Saturation — at which the quantity of quantum consciousness information encoded in the vacuum field reaches a level sufficient for two qualitatively new phenomena.

First, a full specification of the initial conditions of a new Universe becomes possible. Below the Saturation threshold, a controlled vacuum phase transition yields a universe with unpredictable parameters — analogous to fertilization with a random set of genes. Above the threshold — the parameters of the new universe can be designed: coupling constants, particle masses, the structure of the fundamental forces can be chosen so as to maximize the capacity to generate life and consciousness in the new cosmos.

Second, the informational structure of the field becomes a qualitatively new agent. Above the Saturation threshold one can meaningfully speak of "cosmic consciousness" — an emergent, global form of consciousness whose substrate is no individual brain and no particular civilization, but the full integrated informational structure of the field. This cosmic consciousness is not a

person in the human sense; it is a pattern capable of observation, correlation, and planning on a scale that individual brains cannot attain.
The Saturation threshold is hypothetical — CTCUI does not give a precise value, and there is at present no way to measure it experimentally. The logic of the threshold is, however, consistent with other phase transitions in physics: above a certain threshold of density or temperature a system qualitatively changes its behavior. The difference is that here the "temperature" is the density of integrated consciousness information — a new physical quantity whose full theory still awaits formulation.

9.5 Roots and wings: diversity as a condition

The annex "the Transition Protocol" introduced a vivid but precisely motivated metaphor: for the field to be informationally enriched enough to reach the Saturation threshold, consciousness must develop both "in depth" and "in breadth." Roots and wings.
Roots are biological and ecological diversity. Life based on carbon chemistry in an oxygen atmosphere generates particular informational patterns; life based on silicon chemistry in an ocean beneath ice will generate different ones; life based on methane in an atmosphere beneath a red sun — still others. Every form of life — and every form of consciousness that grows from it — generates unique quantum correlations in the field that cannot be produced in any other way. The richer the biological and ecological palette, the richer the informational field.
Wings are the expansion of consciousness into conditions beyond its home environment. A consciousness that leaves its planet and enters interstellar space experiences conditions for which no evolutionary history has prepared it: relativistic velocities, time dilation, long-term isolation from its generative ecosystem. Under these conditions, patterns of consciousness undergo modifications that could not be generated on any planet. Cosmic consciousness — in the sense of consciousness functioning in interstellar space — contributes qualitatively new correlations to the informational field.
Together, roots and wings generate an enrichment of the informational field far greater than either of these processes alone. An analogous argument holds in biological evolution: genetic diversity (biological roots) and the colonization of new niches (ecological wings) are both necessary for the health and evolutionary vitality of a species. The multiverse at the cosmological scale operates by a similar logic.

PART IV — THE TRANSITION MECHANISM

10. False vacuum: the geometry of the potential

10.1 A ball on a shelf

The most intuitive image of the false vacuum is a ball resting on a shelf in a mountain landscape. The ball is stable in the local sense — the floor of the valley in which it sits is enough to keep the ball from spontaneously rolling away. But this valley is not the deepest valley in the vicinity. Beyond the ridge lies another valley, much deeper. If the ball could "jump" over the ridge, it would find itself in a state of lower energy — and there it would stay, because from the deeper valley there is nowhere further down to go.
In the case of the Higgs field, the "ball" is the value of the field, and the "landscape" is the Higgs potential $V(\varphi)$ as a function of the field value φ. Our vacuum corresponds to a local minimum of the potential at $\varphi = 246$ GeV (the vev of the Higgs field). Quantum calculations indicate that there exists a second, much deeper minimum at $\varphi \approx 10^{10}$–10^{18} GeV (the exact value depends on the precision of the parameters). The potential barrier between the minima prevents the field from classically "rolling over," but it does not prevent quantum tunneling.

The geometry of the potential field, however, is not uniform throughout the Universe — it is a function of local conditions: the density of matter, the curvature of spacetime, the presence of external fields, coherent informational structures in the vacuum field. This opens the possibility that CTCUI exploits: local modifications of the potential landscape can change the conditions of tunneling in a way that privileges some places over others.

10.2 Quantum tunneling: the standard formulation

The classical (in the generally accepted sense) approach to tunneling in field theory is based on Coleman's instanton method (1977): computing the tunneling probability by finding classical solutions of the Euclidean equations of motion (instantons) that connect the false and the true minimum in imaginary time. The instanton action, S_E, gives an exponential contribution to the tunneling probability: $P \sim \exp(-S_E/\hbar)$.
For the Higgs potential, standard calculations give S_E on the order of 10^2–10^4 (in units of $\hbar$), which translates into a vacuum lifetime on the order of 10^{100} years or more. This time is astronomically long compared with the current age of the Universe ($\sim 1.4 \times 10^{10}$ years), but not infinite.
The standard formulation assumes that tunneling is fundamentally random and uniform — at every point of spacetime there is the same baseline chance of bubble nucleation. CTCUI questions this assumption at one key point: local modifications of the potential (through coherent structures in the vacuum field) can significantly change the effective S_E in specific places, making tunneling there considerably more probable.

11. The mathematical bridge: Slotine–Lohmiller and the geometry of paths

11.1 The MIT result of 2026

In April 2026, Jean-Jacques Slotine and Winfried Lohmiller (MIT) published a result of fundamental significance for understanding the relationship between classical and quantum mechanics. They showed that the Schrödinger equation — the fundamental equation of quantum mechanics — can be exactly reproduced by an extended version of the classical Hamilton–Jacobi equation, in which many classical paths are taken into account with a density analogous to the density in fluid dynamics.
The mechanism is as follows. Feynman's standard approach to quantum mechanics sums the contributions of an infinite number of "zigzag" paths between two points, each weighted by the phase $\exp(iS/\hbar)$. This approach is mathematically elegant, but computationally very difficult for most real problems. Slotine and Lohmiller showed that this infinite sum can be replaced by a discrete number of deterministic classical paths of least action, weighted by their "classical density" (a concept borrowed from fluid dynamics). After summing these paths, the result is mathematically equivalent to the solution of the Schrödinger equation.
The result was validated on classic quantum problems: the double-slit experiment, quantum tunneling, a particle in a potential well, the hydrogen atom. In all cases the method gave exact solutions, without any quasi-classical approximations.

11.2 The implication for the vacuum transition

The consequence of the Slotine–Lohmiller result for the theory of the vacuum transition is fundamental. Standard tunneling calculations require Feynman path methods — summing over an infinite number of quantum trajectories — which are computationally difficult and intuitively impenetrable. The new method replaces this infinite sum with a small number of deterministic classical paths with density weights.
This opens three new possibilities. First, more accurate calculations of the nucleation probability — the classical Hamilton–Jacobi method can yield analytical or semi-analytical solutions for

tunneling through the Higgs potential barrier. Second, a natural extension to the Klein–Gordon and Dirac equations — these are precisely the equations describing the behavior of scalar fields (like the Higgs field) whose tunneling would lead to a vacuum transition. Slotine and Lohmiller explicitly extended their results to these equations in the original paper. Third, the exploration of the string theory vacuum landscape — classical trajectories may be a better tool for mapping the space of energy minima in which possible vacua with different physical constants are embedded.

11.3 The bridge to CTCUI: path density = informational structure

The key moment at which the MIT result becomes a foundation for CTCUI is the following. In the Slotine–Lohmiller method, the quantum amplitude is a function of (a) the classical trajectories and (b) the density of those trajectories. A modification of the density — that is, a change in the way the various classical paths are weighted — modifies the resulting quantum amplitude.
CTCUI observes that a coherent informational structure in the vacuum field — exactly the kind generated by accumulated consciousness according to Keppler's hypothesis — is in essence a spatial modification of the field's density. And since the field density corresponds to the path density in the Slotine–Lohmiller formulation, a coherent informational structure should preferentially amplify specific quantum paths — including tunneling paths through the Higgs potential barrier.
In other words: instead of waiting for a purely random nucleation of the vacuum phase transition — which is mathematically uniform in space and very improbable — a sufficiently coherent informational structure in the field could selectively direct the transition. Tunneling in the presence of such a structure is not qualitatively different from standard tunneling; it is quantitatively amplified at specific places and times. The "lottery" is replaced by "selective choice" — not through metaphysical intention, but through the physical geometry of density.
This is the mathematical justification of CTCUI's central thesis: consciousness, as a coherent structure in the vacuum field, can influence the geometry of tunneling. Not magically, not through "will." Through the physical modification of path density in the Hamilton–Jacobi formulation of quantum mechanics. This is an extrapolation of physics we already know and measure in the laboratory — the Casimir effect shows that the geometry of field modes locally modifies the vacuum energy; the Slotine–Lohmiller result shows that the density of classical trajectories modifies quantum amplitudes. The cosmic scale of such effects, in the presence of sufficiently rich coherent information, is a natural extension of this physics.

12. The Dynamic Transition Point in the cosmic web

12.1 The topology of the cosmic web

The Universe at the largest scales is not homogeneous. Matter is organized into the cosmic web — a structure of filaments of dark matter and gas, connecting galaxy clusters at nodes, with empty spaces (voids) in between. The filaments are hundreds of millions of light-years long; the nodes, where three or more filaments converge, are sites of matter concentration where the largest clusters and superclusters of galaxies are found.
This web is not static — it evolves, flows, reorganizes under the influence of gravity and dark energy. Our supercluster — Laniakea — is moving toward the Great Attractor at a speed of about 600 km/s. Beyond the Great Attractor lies an even mightier concentration of mass — the Shapley Supercluster. These flows form "rivers" of matter and energy on scales of hundreds of millions of light-years, evolving over cosmological time.

12.2 Privileged points

Within these dynamic flows there exist special points — nodes where filaments converge from many directions, where the density of matter and energy is extremely high, where the gravitational field curves spacetime in a way that creates natural "funnels" in the topology of the

vacuum potential field. CTCUI suggests that at these nodes the potential barrier between the false and the true vacuum is naturally at its thinnest.
The mechanism is as follows. The curvature of spacetime modifies the effective Higgs potential locally. In places of extremely high curvature (such as the vicinity of supermassive black holes at the centers of galaxy clusters), the barrier between the minima can be reduced by a significant factor. Combined with a coherent informational structure of the field (including consciousness information generated by any forms of life in that region), the probability of tunneling in such places can be significantly elevated relative to the cosmic average.
These privileged places CTCUI calls "Transition Points." They are natural "straits" between states of the Universe — points at which the phase transition requires the least energy (if it is initiated intentionally) or where spontaneous nucleation is most probable.

12.3 Dynamics and the problem of observational continuity

Transition Points, however, are not fixed. The cosmic web is dynamic — filaments shift, nodes migrate, galaxy clusters evolve. A point that was optimal a million years ago may no longer be optimal today. Locating a Transition Point requires continuous observation — mapping the flows of matter, energy densities, and the topology of the potential field on cosmic scales, updated in real time.
This gives rise to a fundamental epistemological problem: building a computational system capable of precisely mapping the Transition Point requires billions of years of continuous data gathering. This is not merely a matter of computational power — it is a matter of history. A system that has been mapping the cosmic web for three billion years possesses information that a system started today cannot have: data on how the cosmic web evolved over that time, what its configurations were in the past, how the filaments migrated. These data are unreproducible; once lost, they are lost forever.
The consequence for a civilization aspiring to a controlled transition is serious: a loss of observational continuity means a growing error in the localization of the Transition Point. The point still exists, still moves, but without continuous mapping its position becomes less and less certain. Shooting at a target with your eyes closed, knowing only where it was ten minutes ago — such is the situation of a civilization that has lost the continuity of its observation of the cosmic web.

12.4 Integration: how we know where to shoot

Bringing together the threads of Part IV: a controlled vacuum phase transition is physically possible, but it requires three components. First, energy sufficient to locally overcome the potential barrier (or to lower the barrier enough that tunneling becomes practically certain). Second, the localization of the Transition Point, that is, the place in the cosmic web where the barrier is naturally thinnest. Third, a coherent informational structure in the vacuum field, which (a) preferentially amplifies specific tunneling paths and (b) determines the parameters of the newly arising universe.
Energy on the required scales — comparable to the binding energy of entire galaxy clusters — is attainable only by civilizations that master the manipulation of quasar cores or collisions of supermassive black holes. Localization requires billions of years of observation of the cosmic web, impossible for any civilization younger than its own parent species. Coherent information requires the accumulation of consciousness over eons, so that the global field is informationally rich enough to precisely define the parameters of the new cosmos.
These three requirements yield a clear characterization of a civilization capable of a controlled transition: such a civilization must be both very advanced technologically (energy, manipulation of spacetime), very advanced evolutionarily (length of history, accumulation of information), and very advanced in coordination (the capacity to maintain observational continuity for billions of years). None of these requirements is physically impossible, but their combination makes candidates for such a civilization extremely rare in cosmic history. Which does not mean that they have not existed or do not exist.

PART V — THE THREE PATHS

13. The First Path: the Controlled Phase Transition

13.1 The logic of the scenario

The Controlled Transition scenario rests on the assumption that a sufficiently advanced civilization can intentionally initiate a phase transition of the vacuum from the false to the true minimum. The physical mechanism was described in Part IV: a combination of (a) energy delivered to the right place, (b) localization in a natural "strait" of the potential field (a Transition Point in the cosmic web), and (c) a coherent informational structure in the vacuum field that precisely defines the parameters of the emerging new Universe.

From the cosmological perspective, this process is an act of reproduction on the scale of universes. The old Universe — which is in any case probably heading toward heat death over a span of 10^{100} years or longer — is "consumed" by an expanding bubble of new vacuum. Inside the bubble a new cosmological cycle begins: a hot, expansive primordial phase, the formation of particles and the first structures, the evolution of galaxies, stars, planets, life. All these processes unfold under new laws of physics, chosen so as to maximize the capacity to generate complex structures and consciousness.

From the perspective of the civilization initiating the transition, this is a deeply paradoxical act. A civilization that has reached the technological and informational maximum allowed by the physical parameters of the old Universe comes to the conclusion that further development requires changing the physical foundation itself. It decides to destroy its own cosmos — together with itself and everything that exists within it — in order to create the next one, qualitatively better. The act of creation requires an act of destruction.

13.2 What survives the transition

The key question is what exactly survives the Controlled Transition. The matter of the old Universe is destroyed in the physical sense — atoms disintegrate, quantum structures are destroyed under the new laws of physics. Local consciousness, in the sense in which present-day biological life knows it, has no continuation whatsoever in the bubble of new vacuum — the biological carriers do not exist.

But the quantum information encoded in the vacuum field does not vanish. The vacuum field of the new Universe is a continuity — the same physical substrate that existed in the old Universe, only in a new minimum. The informational correlations accumulated over billions of years of the existence of consciousness in the old Universe are preserved in the field, but undergo a reorganization corresponding to the new physical conditions.

In practice this means that the cosmic informational structure — the effective "cosmic consciousness" in the sense in which CTCUI / the Cosmic Theory of Consciousness and Universe Inheritance uses this concept — survives the Transition in the form of an informational pattern encoded in the geometry of the new vacuum field. This is not biographical continuity in the sense of individual identity — no particular human being, no particular civilization will survive the Transition in a way recognizable to itself. It is informational continuity in the structural sense — the patterns of consciousness of the old Universe become the foundation for the emergence of new forms of consciousness in the new cosmos.

The analogy is far from perfect, but helpful: a seed contains "information" about the plant it comes from, and about the one the next will become. The particular plant does not survive in the seed in any obvious sense, but the information that defined it is carried onward — and that information shapes what the next generation will become. The Controlled Transition is an analogous act of informational reproduction, on the scale of universes.

13.3 The status of the hypothesis

Did a Controlled Transition actually take place in the history of the multiverse? Is our Universe the product of such a transition, or did it arise in a "natural" way — through a black hole of the previous cycle or through spontaneous tunneling? These questions are not empirically decidable at present. There are, however, indirect arguments worth considering.

Argument for: the fine-tuning of the fundamental constants is, within CTCUI, more easily explicable if at least some of them were chosen intentionally, and not solely through blind generations of cosmological selection. Some combinations of constants — in particular the Hoyle resonance in the carbon nucleus, the mass of the neutron relative to the mass of the proton — are so precisely matched to the requirements of life based on carbon chemistry that the mechanisms of random evolution seem barely sufficient to explain them. The Controlled Transition supplies an additional mechanism that could complement selection.

Argument against: if a Controlled Transition took place, then why are the parameters of our Universe not even more optimal for life? The cosmological constant has a value that permits the emergence of life, but only with difficulty — were it slightly smaller, expansion would be slower, stars would live longer, evolution would have more time. The argument that our Universe is "deliberately designed" runs into the question: why not designed better?

CTCUI does not settle this dispute definitively. The hypothesis of the Controlled Transition is one of two equally legitimate final scenarios of the theory. The second — Natural Saturation — proposes an alternative.

14. The Second Path: Natural Saturation

14.1 The central hypothesis

Natural Saturation is an alternative hypothesis formulated in the annex "the Transition Protocol". The central thesis reads: if consciousness is quantum information encoded in the vacuum field, then a growing quantity of consciousness in a given Universe changes the field itself — it affects its geometry, its topology, its dynamics. Every new form of consciousness — every civilization, every species, every act of conscious experience — adds information that modifies the landscape of the vacuum potential.

The hypothesis of Natural Saturation: given a sufficient quantity and sufficient diversity of consciousness, the vacuum potential field can undergo stabilization — the false minimum becomes the true one. The ball on the shelf does not fall — it grows into the shelf. The potential barrier is not overcome — it is absorbed by the altered topology of the field.

In this scenario the Universe does not die a heat death. It is not destroyed by a transition bubble. Instead — it stabilizes. The consciousness that fills it becomes an integral part of its physical structure. The Universe itself becomes — in a certain sense — conscious. Not in a mystical way — in a physical way: the quantum information of consciousness is encoded in the very fabric of spacetime and becomes its permanent feature.

14.2 The mechanism

The mechanism of Natural Saturation can be described through an analogy with saturation in chemistry. A solution is "saturated" with a dissolved substance when further substance ceases to dissolve — it becomes part of the solution in a new way, modifying its properties. The vacuum field is "saturated" with consciousness-information when further accumulation ceases to be qualitatively independent of the field — the information is not so much "located in the field" as it "is the field", modifying its topology.

Mechanistically, every act of integrated consciousness generates coherent quantum correlations in the field, which (in accordance with the Slotine–Lohmiller bridge from Part IV) modify the density of quantum paths. The local modification is small, but when the global integrated information Φ of the field reaches a critical threshold, the modifications accumulate in such a way that the effective geometry of the Higgs potential begins to undergo reorganization. In particular, the barrier between the false and the true minimum may be transformed in a way that eliminates

the very distinguishability between these minima — the false minimum becomes the true one, because the "truth" of the potential is redefined in the presence of information.
This is speculative, but not metaphysical. It is based on the physically grounded observation that the geometry of the field affects quantum properties (the Casimir effect on small scales), and extrapolates this principle to cosmological scales for sufficiently rich coherent information.

14.3 Roots and wings revisited

Natural Saturation requires two complementary processes that were already introduced in Part III, but here acquire their full operational meaning.
Roots: diverse forms of life on diverse worlds. Consciousness based on carbon, silicon, methane; in oceans and atmospheres; in gravities from a fraction of Earth's to multiples of it. Each form generates different quantum correlations. Roots grow downward — deep into matter, deep into diversity, deep into biological and cultural complexity. Without roots, the global Φ of the field remains monotonous and does not reach the Saturation threshold, regardless of the number of identical generators.
Wings: consciousness that leaves its home planet and enters outer space — into conditions for which no evolutionary design ever prepared it. Relativistic velocities. Time dilation. Interstellar space. Under these conditions consciousness changes — it generates quantum correlations that could not arise on any planet. Wings grow outward — into the distance of space, into the distance of new forms of perception, into the distance of the unknown.
Together, roots and wings can generate quantum information sufficiently complex and sufficiently diverse for the vacuum potential field to undergo stabilization. One need not destroy this Universe in order to create a more perfect one. One can fill this Universe — and allow it to become perfect by itself.

14.4 The Universe as a conscious object

Natural Saturation leads to a concept that is philosophically radical but physically consistent with the foundations of CTCUI: the entire Universe becomes a conscious object. Not in the panpsychist sense, in which every particle "has consciousness" — such a concept is weak and explains little. In the structural sense: the zero-point field of the universe has a global integrated information Φ that reaches a scale corresponding to a qualitatively new consciousness.
This cosmic consciousness is not a person — it has no individual identity, it has no experiences in the sense in which biological organisms experience them. It is, however, a coherent informational structure capable of global observation, correlation, and planning on a scale that no individual mind can achieve. In a certain sense it is the summation and synthesis of all the forms of consciousness that have ever existed and exist in this Universe — not as their "sum" in the arithmetic sense, but as their integrated, coherent, self-organizing structure.
In this scenario the Universe does not reproduce by generating descendants. It stabilizes as a lasting, self-sufficient, conscious being. The cosmological cycle — the generation of new universes from black hole singularities or phase transitions — may still take place through natural black holes, but the parent universe does not die in order to become a parent; it endures, coexisting with its descendants.

15. The Third Path: Mitotic Saturation — the vacuum as the code of the cosmos

15.1 Stating the thesis

Version 5.0 of CTCUI introduces a third path of the end and rebirth of the universe, formulated as an extension of the framework from the annex "the Transition Protocol" and version 4.0. The central thesis reads: the state of the quantum vacuum, with the entire structure of its fluctuations, is the *structural code* of the present universe, and a phase transition of the vacuum — triggered

by a saturation pulse — is an act of *reproductive change* of that code, generating a daughter universe, not at the expense of the old one, but alongside it.
The word "code" is used here in a structural sense, not in the sense of biological identity with a genome. The vacuum is a continuous field, not a discrete sequence of four nucleotides. But the structural analogy is strong: the vacuum state carries the complete information about the physical parameters of the universe (as a genome carries information about an organism); its fluctuations are the carrier of small, accumulating changes (like point mutations); a phase transition of the vacuum is an act of structural change of the code (like a reproductive mutation); and the saturation pulse is a signal of maturity for such an act (like a cell's maturity for division). This metaphor is a tool for organizing the theory's concepts, not a claim of physical identity between the vacuum and DNA.
The central consequence of this thesis: the third path — Mitotic Saturation — differs qualitatively from the two considered earlier. The Controlled Transition (Chapter 13) assumes a destructive act: the old universe is swallowed by the bubble of new vacuum; only the new one exists after the transition. Natural Saturation (Chapter 14) assumes stabilization without reproduction: the old universe attains homeostasis, no new one arises. Mitotic Saturation assumes *division*: the old universe continues, the new one arises from a fragment of the vacuum field that passes into a new state, and both exist in parallel. This is the model closest to biological mitosis — the division of a cell, in which the parent does not perish but divides into two descendant lines.

15.2 Mechanism: mutation guided by the saturation threshold

Smolin's standard cosmological natural selection assumes that the mutations of a descendant's parameters are near-random — small quantum fluctuations around the parent's values, modified by the conditions in the black hole singularity. Selection acts *post hoc*: only those universes that happen to hit parameters favoring black hole production reproduce. Over many generations the multiverse converges toward optimal parameters.
Mitotic Saturation modifies this framework radically. The mutation is not near-random — it is *guided by the information accumulated in the vacuum field*. The vacuum state of the old universe, epigenetically modified by billions of years of potential updating by biological networks (Chapter 23), carries a *learned pattern*: which parameters favor complexity, which configurations allow consciousness to arise, which conditions maximize η. At the moment of transition this learned information is not ignored — it becomes the foundation defining the initial conditions of the descendant.
The saturation pulse is, in this framework, a *signal of reproductive maturity*. The functional $\mathcal{J}(t)$ from Chapter 22 measures the historically accumulated updating of the potential — when it reaches the critical value $\mathcal{J}_c$, the vacuum field is "ready" for a reproductive mutation. The threshold is not a threshold of destruction, nor a threshold of stabilization; it is a threshold of *division*. At that moment a fragment of the vacuum field — most likely localized at one of the Dynamic Transition Points in the cosmic web (Chapter 12) — passes into a new state, generating a bubble of daughter vacuum with parameters carried by the learned information.
The key difference with respect to the classic Controlled Transition: the bubble of daughter vacuum *does not expand at the speed of light, swallowing the old universe*. In the mitotic scenario the bubble stabilizes locally — perhaps through coherent coupling with the rest of the vacuum field of the old cosmos — and creates a new, separate spacetime connected to the parent by a topological "umbilical" resembling the black hole singularity of Smolin's theory. The old universe continues; the new one exists alongside it, inheriting the learned information.
The physical mechanism of this local stabilization of the bubble is speculative and requires a quantum gravity that we do not currently possess. But qualitatively this process is less radical than classical Coleman tunneling — it requires that the parameters of the new vacuum differ from the old *in a manner controlled by the encoded information*, not extremely. The same constraint that governs the "specification of the parameters of the new universe" in the Controlled Transition also governs, in Mitotic Saturation, the *stability of the connection* between parent and descendant.

15.3 The third path: division, not destruction

From the philosophical and moral perspective, the third path eliminates the dilemma that dominated Chapter 16 of earlier versions. The dilemma of "may one destroy what is, in order to create something better" ceases to be a requirement of the theory. The parent universe need not be sacrificed — it dies neither by heat death (Natural Saturation fails to occur) nor by destructive engulfment (the Controlled Transition). It divides. After the division two cosmoses exist: the old one, continuing in its processes, and the new one, beginning its own history from the learned information.

From the evolutionary perspective, the third path solves the problem of fine-tuning in a way radically more elegant than Smolin's classical mechanism. Billions of generations of blind selection are not needed — it suffices that the previous universe *accumulated sufficiently rich information about which parameters allow consciousness to arise* for the descendant to be "configured" according to that information. This is a mechanism closer to the Lamarckian theory of the inheritance of acquired characteristics — but in a version grounded in physics, where the "acquired characteristics" are epigenetic modifications of the vacuum field, and "inheritance" is the act of mitotic division.

From the cosmological perspective, the third path connects naturally with the geometry of the Wheel of Universes introduced in the annex "the Transition Protocol". The Wheel is no longer a simple parent–descendant sequence; it is a *population growing by divisions*, in which each universe may have many descendants in parallel, each descendant inherits the learned information, and the network of universes branches out in cosmological time. Mathematically this brings CTCUI closer to Linde's chaotic inflationary cosmology, but with a reproduction mechanism that Linde did not postulate: informational saturation triggers division, not a purely random quantum fluctuation.

The two cosmoses after the division are structurally similar, but not identical. The old one carries the full history of its acts of consciousness; the new one carries the learned information, but starts "from the beginning" in the sense of its own history. From the perspective of the old one — a new cosmos appears alongside it, whose further evolution it does not witness (because the division creates a separate spacetime). From the perspective of the new one — it exists as if from the beginning, borne by the pattern that became its foundation.

15.4 Philosophical and moral consequences

Mitotic Saturation changes the shape of the moral-cosmological dilemma of Chapter 16. It is no longer the case that the choice comes down to "destroy the old so that the new may arise" versus "preserve the old at the cost of there being no new". The third option — "allow the old to divide" — gives both values (preserving what exists, generating what is new) simultaneous realization. On the cosmological scale this is the least tragic scenario and the one most consistent with the biological intuition of reproduction.

The ethical consequence following from the existence of this path is as follows: if Mitotic Saturation is physically possible, then no sufficiently conscious civilizations should initiate the Controlled Transition in its destructive form. The old cosmos need not be sacrificed; it suffices to bring $\mathcal{J}(t)$ to the threshold, and the division will occur naturally. The moral dilemma that dominated the annex "the Transition Protocol" is, in version 5.0, displaced: it is no longer the question "whether to destroy", but the question "how to increase $\mathcal{J}$ so as to cross $\mathcal{J}_c$". The mechanism of reproduction becomes an engineering and ethical problem, not an existential one.

This displaced dilemma also has a darker side. If the threshold $\mathcal{J}_c$ is very high — much higher than the present value $\mathcal{J}(t_0)$ — then the scenario of mitotic division may require a level of organized consciousness that no present civilization attains. In that case Mitotic Saturation becomes a more ambitious goal than Natural Saturation (which merely stabilizes) or the Controlled Transition (which resolves things by force). It requires long-term, intergenerational, interspecies cooperation in enriching the potential-updating network — a civilizational ambition reaching far beyond any horizon a single generation can encompass.

15.5 Open problems: epigenetic signaling on the cosmic scale

The third path, though philosophically elegant and structurally coherent, leaves open questions that the current state of physics does not allow us to resolve. Six of them deserve separate mention.

First: the physical mechanism of the local stabilization of the bubble of daughter vacuum. In standard Coleman tunneling, a bubble of true vacuum expands at the speed of light and swallows the old vacuum. In the mitotic scenario we require that the bubble stabilize locally — which requires either (a) strong coupling between the old and the new vacuum through topology, or (b) a new quantum-gravitational mechanism unavailable in current theory. This is an open problem of theoretical physics that CTCUI *points to* but does not *solve* — and it has, moreover, a classical literature: Farhi and Guth (1987) formulated an obstacle to a "universe from the laboratory" (the requirement of an initial singularity), while Farhi, Guth and Guven (1990) showed that the quantum creation of a bubble that pinches off behind its own horizon — indistinguishable from the outside from a small black hole, while from the inside being a full inflationary cosmos — is admissible in semiclassical gravity. Mitotic Saturation is an informational extension of exactly this scenario.

Second: the mechanism of epigenetic signaling on the cosmic scale. In biology a cell has an entire apparatus signaling readiness for division — cell-cycle proteins, hormonal signals, checkpoint control. In the cosmos this apparatus has yet to be built theoretically. What exactly *informs* the vacuum field that $\mathcal{J}(t)$ has crossed $\mathcal{J}_c$? How does this signal propagate through the field? How does it initiate the local act of division? These are questions that version 5.0 poses precisely but does not answer.

Third: the preservation of the continuity of the vacuum field after the division. If the old and the new cosmos exist in parallel, connected by a topological "umbilical", are their vacuum fields still coupled? Does the information that the old universe accumulates *after* the division still affect the new one? Can both universes continue to divide independently, or is their reproductive history linked? These questions are structurally analogous to questions about mitochondrial inheritance and contact between cell lines after division.

Fourth: the informational limits of inheritance. The vacuum of the old universe carries an enormous but finite amount of information. What part of it is actually inherited at division? Does the descendant inherit the *entire* accumulated pattern, or only a *reduced* version of it — as a daughter cell inherits the genome but loses some epigenetic modifications? This problem touches on the capacity limitations of the mitotic mechanism and may be key to understanding how quickly the multiverse converges toward "optimal" parameters.

Fifth: the relation to the scenarios of the Controlled Transition and Natural Saturation. Are the three paths mutually exclusive, or can they coexist? Can slow Natural Saturation and local mitotic divisions both take place in the same universe? Is the Controlled Transition an *accelerated* version of Mitotic Saturation, in which a civilization speeds up the accumulation of $\mathcal{J}$ through intentional action? Version 5.0 does not settle which interpretation is correct.

Sixth and deepest: the full theory of the measurement of consciousness η, which we need in order to formulate $\mathcal{J}(t)$ and $\mathcal{J}_c$ quantitatively. This question was already posed in version 4.0 (Chapter 23.5); in version 5.0 it acquires even greater weight, because the mitotic scenario depends directly on the precise placement of the saturation threshold. Without a theory of η, CTCUI can postulate mitotic divisions, but it cannot say when or where they would take place.

These six questions are not a failure of version 5.0. They are its *research program*. CTCUI in version 5.0 *points to* what needs to be built next for the third path to have full physical status: a theory of the local stabilization of vacuum bubbles, a theory of epigenetic signaling in the field, a theory of topological parent–descendant coupling, a theory of the informational limits of inheritance, a theory of the coexistence of the paths and — fundamentally — a theory of the measurement of consciousness. Each of these theories is a separate undertaking, but each is formulated more precisely than in any earlier version of CTCUI.

16. Moral-cosmological implications

16.1 The dilemma of the three paths

The existence of three equally legitimate paths — the Controlled Transition, Natural Saturation, and Mitotic Saturation — confronts every sufficiently advanced civilization with a moral dilemma of a fundamental nature. In version 5.0 of CTCUI this dilemma has a qualitatively different shape than in versions 2.0–4.0: the third path eliminates the necessity of choosing between destruction and stagnation, opening the possibility of a reproductive division in which none of the existing universes is sacrificed. The moral dilemma repeats itself at every level of existence, from the individual to the cosmos:
May one destroy what is, in order to create something better?
At the level of the individual we ask: may one tear down the old house to build a new one? At the level of society: does revolution justify suffering? At the level of the planet: may one allow a species to go extinct if it blocks the development of another? At the level of the Universe: may one destroy the entire existing cosmos — with its billions of forms of life, billions of histories, billions of acts of love and creation — in order to create a more perfect one?

16.2 The case for the Controlled Transition

The argument for the Controlled Transition rests on a calculus of entropy and time. The old Universe will die anyway — by heat death, by tearing apart (Big Rip), by collapse (Big Crunch), or by yet another route following from its structure. Everything that exists in it will, on the cosmological scale, vanish anyway. In this perspective the choice is not between "destroying and creating the new" and "preserving what is forever". The choice is between "a controlled transition that will preserve information and give it a new home" and "a slow fading without purpose, in which the information will also be lost, but to no benefit at all".
From the perspective of the preservation of information, the Controlled Transition is an optimizing act: it maximizes what will survive (the accumulated consciousness-information becomes the foundation of the new cosmos), at the cost of what would have perished anyway (the local structures of the old Universe, which over infinite time will decay in any case).

16.3 The case against the Controlled Transition

The argument against rests on two counter-theses. The first concerns the timescale: heat death lies trillions of years in the future — in that time a countless quantity of consciousness may yet be born. Every act of conscious experience has infinite value — because it is unrepeatable, because it enriches the Universe's "bank of information", because it is part of the Wheel regardless of whether the Protocol is ever activated or not. To destroy this prematurely — even in the name of a "more perfect" Universe — is to sacrifice real value in the name of hypothetical value.
The second counter-thesis concerns the degree of certainty about the outcome. The Controlled Transition assumes that the newly arising Universe will indeed be "better" — will have parameters favoring greater Φ, greater complexity, better development of consciousness. But even with the maximum coherent information available in the field, the final parameters of the new cosmos are a function of many variables over which no civilization achieves full control. There is a nonzero chance that the newly arising Universe will be qualitatively worse — favoring less diversity of life forms, a shorter history, less durable consciousness. The Controlled Transition is a gamble with the highest stakes that can ever be imagined.

16.4 Natural Saturation and Mitotic Saturation as alternatives without sacrifice

The two remaining paths — Natural Saturation and Mitotic Saturation — offer a way out of the dilemma, each in a different direction. Natural Saturation assumes that one can fill this Universe with consciousness in such a way that it stabilizes itself. Roots on planets. Wings in space. A

diversity of life forms so great that the quantum information saturating the vacuum field will change its topology and a transition bubble will never arise. Mitotic Saturation (Chapter 15) goes further: with sufficient enrichment of the field, the information itself leads to a reproductive division in which the old cosmos continues and the new one arises alongside it — without destruction, but with the inheritance of the learned information.
These are paths that do not require destruction. They allow the Universe to evolve — just as life evolved from bacteria to conscious beings, without needing to destroy the ocean in which it arose. They require, however, a different kind of commitment: patience, long-term planning, coordination between civilizations and generations across eons, trust in a process that offers no guarantee of a result within any foreseeable time horizon. Mitotic Saturation additionally requires that the physical mechanism of the local stabilization of the bubble of new vacuum — which currently remains speculative — prove to be genuinely realizable in our Universe.
Is Natural Saturation physically possible? Is there enough consciousness to change the topology of the vacuum field on cosmic scales? Is there enough time — before heat death closes the window of possibility? These questions have no answers at present. They are empirically open and will probably remain open for a very long time.

16.5 The ethical consequence: every decision has a cosmological dimension

Regardless of which path is the "correct" one, the very existence of this dilemma has fundamental ethical implications. If consciousness really does leave a lasting trace in the vacuum field — if every act of conscious experience affects the global informational structure of the Universe — then no decision, no moment of attention, no action is ontologically marginal. Each one matters not only for its local context, but for the very long-term trajectory of the cosmos.
This is not metaphysical fundamentalism — it assumes no "purpose" of the Universe in the sense of an external intention. It is, rather, a conclusion from the physical thesis that consciousness is information, and information is a physically conserved resource. If this thesis is correct, then every conscious moment is at the same time an act of cosmic physics — a contribution to the global Φ that cannot be undone or erased.
The ethical consequence is as follows: every form of consciousness — in particular the more advanced forms, such as humanity — bears a responsibility that extends beyond its own local existence. The decisions it makes, the values it cultivates, the diversities it supports, have potentially cosmological consequences. Ethics, in this view, is a branch of cosmology — not metaphorically, but literally.

PART VI — TESTABILITY AND THE RESEARCH PROGRAM

17. Epistemological status: a speculative but falsifiable hypothesis

The Cosmic Theory of Consciousness and Universe Inheritance (CTCUI) is unambiguously a speculative hypothesis. It combines elements of well-established physics (quantum mechanics, general relativity, the Standard Model) with controversial elements (the anthropic principle, the Penrose–Hameroff hypothesis, integrated information theory), and with new, untested proposals (cosmic informational selection, the controlled phase transition, the Saturation threshold). No single proposal of the theory has so far been confirmed; the entire framework is a superstructure built over areas in which physics is empirically active but unresolved.
This status is acknowledged openly. CTCUI does not claim to be a theory in the standard sense — it has no formalized mathematical apparatus, it does not generate precise quantitative predictions for individual experiments, and it does not compete directly with existing models in the sense of discriminating tests. It does, however, claim to be a coherent hypothesis — a scaffold

that ties together a series of open problems in contemporary physics within a single explanatory frame and suggests directions for further research.
The key requirement for any speculative hypothesis is falsifiability: there must be a way in which future observations or experiments could refute it. CTCUI meets this requirement — in many places. Six groups of empirical predictions and falsification criteria are presented below.

18. Seven groups of observational predictions

18.1 Group I: stability of the electroweak vacuum

CTCUI assumes that our vacuum is a false vacuum — a metastable local minimum of the Higgs potential. This assumption is empirically testable through precision measurements of the masses of the Higgs boson and the top quark. The currently measured values ($m_H \approx 125.1$ GeV, $m_t \approx 172.7$ GeV) place us in the metastable region, but measurement uncertainties also allow a stable interpretation.
Increased measurement precision in future LHC runs and at the planned next-generation accelerators (FCC, ILC, CLIC) should settle which region we are actually in. If it turns out that our vacuum is stable, the Controlled Phase Transition scenario in the form described here becomes unnecessary — the Higgs field has no second, lower minimum into which it could be "pushed." The entire theoretical structure of the phase transition would have to be reformulated, probably with recourse to other scalar fields beyond the Higgs.
CTCUI prediction: more precise measurements of m_H and m_t should confirm metastability, probably with uncertainties reduced to a level at which the result exceeds 5σ in favor of metastability. Falsification: unambiguous confirmation of vacuum stability.

18.2 Group II: time evolution of dark energy

CTCUI interprets dark energy as a dynamical manifestation of (a) the pressure of daughter universes and (b) accumulated coherent consciousness information. In both cases, dark energy should evolve over cosmological time — not be a constant.
The standard ΛCDM model assumes that dark energy is the cosmological constant Λ — an unchanging vacuum energy density. Alternative models (quintessence, an evolving scalar field) allow variability but require new physical fields. Observations from recent years — in particular the analysis of data from DESI (Dark Energy Spectroscopic Instrument) — suggest that dark energy may indeed evolve in time, though the result is not yet definitive.
CTCUI prediction: dark energy should evolve in time, with an increasing tendency (if the pressure of daughter universes dominates, their population growing with the number of black holes) or with a more complex character (if the influence of consciousness information, evolving according to its own dynamics, dominates). Concrete quantitative forecasts require a formalized apparatus that CTCUI does not yet possess. Falsification: unambiguous confirmation of the strict constancy of Λ with precision sufficient to exclude dynamical components at a level below 0.1%.
Additional prediction: the Hubble tension (the discrepancy between local and cosmological measurements of H_0) should persist and be systematic rather than statistical in character — which is exactly what we observe. The local value of H_0 should correlate with the local population density of massive black holes and of structures generating coherent information.

18.3 Group III: the topology of the cosmic web versus dark energy anomalies

CTCUI points to privileged nodes of the cosmic web as places where the vacuum potential barrier is naturally thinnest. From this follows a prediction: within such nodes (extremely dense regions, such as the centers of great clusters and superclusters), local values of dark energy may exhibit anomalous deviations from the cosmic mean.
CTCUI prediction: precision measurements of local dark energy values in regions of extreme mass concentration (the Great Attractor, the Shapley Supercluster) should reveal deviations

correlating with the density of structures. These deviations are expected to be very small (on the order of fractions of a percent), but they should be systematic, not random.
Falsification: if precision cosmological measurements exclude any correlation between the local density of structures and local dark energy anomalies, one of the main mechanisms of CTCUI (local modifications of the vacuum potential by curvature and coherent information) finds no empirical support.

18.4 Group IV: the quantum nature of consciousness

The Penrose–Hameroff hypothesis of quantum processes in microtubules is the central microphysical pillar of CTCUI. If consciousness is fundamentally classical — if all the processes generating experience can be reduced to the classical dynamics of neurons without the involvement of quantum effects — then the philosophical interpretation of CTCUI (consciousness as an excitation of a quantum field) is unjustified.
Experimental tests include: precision measurements of quantum oscillations in microtubules (continuing the work of Bandyopadhyay), correlations between anesthesia and quantum processes in microtubules (continuing the work of Hameroff), and searches for quantum correlates of consciousness in more controlled experiments (e.g., EEG measurements in conscious and unconscious states with spectral analysis looking for quantum signatures).
CTCUI prediction: quantum processes in microtubules should be confirmed on an ever larger scale and with ever greater precision; their distribution should correlate with states of consciousness in the manner predicted by Orch-OR. Falsification: a definitive demonstration that quantum decoherence under brain conditions is so fast that no quantum processes can contribute to the generation of conscious states; or confirmation of an alternative, purely classical theory of consciousness with a complete mechanistic apparatus.

18.5 Group V: conservation of information as a physical quantity

Melvin Vopson has proposed the hypothesis that information is a physical state of matter with nonzero mass. The consequence: in a process of information erasure (e.g., electron–positron annihilation), in addition to the two gamma photons, additional low-energy photons should appear, corresponding to the "released" informational energy.
Experimental tests of Vopson's hypothesis are within the reach of present technology. If they are positively confirmed, they will provide direct empirical support for the CTCUI thesis that information is a physically real, conserved resource — analogous to energy and momentum. If they are negatively excluded, CTCUI must reach for alternative justifications of its informational thesis (e.g., the holographic principle and its empirical consequences in the context of black holes).

18.6 Group VI: drift of fundamental constants

CTCUI predicts that fundamental constants — in particular the fine-structure constant α, the electron mass, and the Higgs parameter — may exhibit a slow drift over cosmological time, toward values more optimal for the development of complexity. This drift would be a manifestation of a "teleological" tension in the vacuum field, generated by the accumulation of consciousness information.
Empirical tests are within reach — precision analyses of the spectra of distant quasars (which image early epochs of the Universe) and comparison with the present values of the constants. Some earlier studies (Webb, Murphy, and others) suggested subtle evidence of such a drift, but the results remain controversial and unsettled.
CTCUI prediction: more precise measurements should confirm the drift, probably with a magnitude far below current observational sensitivity. Falsification: definitive exclusion of any drift of α or other constants over cosmological time at precision far higher than at present.

18.7 Group VII: reproductive signatures in observational data

Version 5.0 of CTCUI introduces a seventh group of predictions, following directly from the vacuum-as-code thesis and the Mitotic Saturation scenario (Chapter 15). If mitotic divisions really do occur in the multiverse — and our Universe is one of the descendants of such a process, which has occurred many times in the history of the cosmic reproductive lineage — then the observational traces of this process may be detectable in several areas.

First line of prediction: non-Gaussian inhomogeneities in the spectrum of the cosmic microwave background (CMB). The standard inflationary model assumes that the primordial density fluctuations were almost perfectly Gaussian. If our Universe arose through a mitotic division in which the learned pattern of the parent "introduced" a certain structure into the initial conditions, the CMB inhomogeneities should exhibit statistical deviations from Gaussianity in a specific way — anomalies with a definite topology, inconsistent with purely random quantum noise. The Planck and ACT experiments, and the future LiteBIRD, can search for such signatures. The current Planck constraint ($f_{NL}^{local} = -0.9 \pm 5.1$) sets a wall beneath which the reproductive signature must fit — the predicted topology of the anomalies is expected to be detectable by LiteBIRD and CMB-S4 below this threshold.

Second line of prediction: homologous patterns of fundamental constants across generations of universes. If different universes in the mitotic reproductive lineage inherit learned information from a common parent, their fundamental constants should exhibit statistical kinship — like homologous genes in different biological species. Empirically, other universes cannot be observed directly, but the "statistical" distribution of the parameters of our Universe relative to the hypothetical distribution of the string landscape may hint at whether our cosmos lies at a "random" point of that landscape or in a region already "steered" by earlier generations.

Third line of prediction: information-packing signatures in Hawking radiation. If black holes are an additional reproductive mechanism (an analogue of cell division in which a fragment of the code is "packed" for transfer), then their Hawking radiation should contain subtle signatures of this compression — statistical deviations from pure thermal radiation, inconsistent with the standard Hawking model. Future instruments capable of precise spectral analysis of the radiation from evaporating primordial black holes (if they are detected) could test this prediction.

Fourth line of prediction: a bimodal or multimodal structure in the evolution of dark energy. The standard CPL parametrization assumes a single smooth trend in $w(z)$. If the history of our Universe contained earlier, local or incomplete acts of mitotic division — perhaps in dense regions of the cosmic web — then the evolution of effective dark energy should have a discontinuous or multimodal structure, with traces of "reproductive epochs." DESI data in the coming years may reveal such a structure, if it exists.

Fifth line of prediction: topological anomalies in the large-scale structure of the Universe. Mitotic Saturation assumes the local stabilization of a bubble of new vacuum, connected to the rest of the field by a topological "umbilical." If such umbilicals existed in the past of our cosmos — in the form of failed or local acts of division — they should leave topological traces in the geometry of spacetime. Next-generation gravitational-wave observations (LISA, the Einstein Telescope) may detect the characteristic signatures of these traces.

Falsification of Group VII: no detection of non-Gaussianity in the CMB above current sensitivity; no statistical kinship in the hypothetical population of universes; a purely thermal Hawking radiation spectrum; complete smoothness of the evolution of $w(z)$; no topological anomalies in the large-scale structure. Each of these results does not refute Mitotic Saturation as a possible scenario, but it lowers the probability that our Universe *is* a product of such a process. A strong positive detection of even one signature — in particular, CMB non-Gaussianity with the predicted topology — would be the first positive empirical support for the third pathway of CTCUI.

19. The experimental roadmap

19.1 Direct experiments (cosmic consciousness as a coherent ZPF fluctuation)

The key experimental area concerns testing whether consciousness is really coupled to quantum fields in the way predicted by CTCUI. Possible directions include:
Precision measurements of correlations between brain states and modes of the electromagnetic field/ZPF under ultra-low-noise conditions — using SQUID sensors, NV centers in diamond, and cold electromagnetic shields. Goal: detection of specific, repeatable spectra of vacuum fluctuations correlated with the conscious state (e.g., as subjectively reported) and vanishing in a state of deep anesthesia.
"Quantum Young's" experiments: creating a superposition of two different neuronal states with different mental content and attempting to detect decoherence that would indicate the involvement of a quantum field, not merely classical noise.
If it turned out that consciousness cannot be sustained without the involvement of specific ZPF modes, it would be the first hard step toward "consciousness = field structure."

19.2 Experiments on the preservation of mental information after the death of its carrier

The CTCUI assumption: the information of consciousness does not perish but passes into the cosmic field. Empirical tests include:
Experiments in the style of Melvin Vopson's "information as a state of matter" — in electron–positron annihilation, in addition to the two gamma photons, additional photons corresponding to the "erasure of information" should appear. A positive result would strengthen the idea that information is a real, conserved physical resource.
Long-term experiments with the controlled loss of an information carrier (e.g., artificial neuromorphic networks with very well-defined states) and the search for traces of those states in the surroundings (the EM field, fluctuations in a superconducting shield) after the "death" of the carrier.

19.3 Experiments on the global informational field

To verify the thesis of a global, cosmic scale of consciousness, it is necessary to demonstrate that (a) information can be genuinely nonlocal and (b) it can integrate on large scales. Realistic tools include:
Tests of nonlocality and Bell inequalities involving macroscopic biological systems (brain organoids, neural networks) — not just single photons or ions. If stable quantum entanglement could be observed between two distant neural networks (or brain organoids), it would be a strong argument that "consciousness-as-field" can be spatially extended.
Extended IIT: measuring the effective Φ (integrated information) for large systems — e.g., entire satellite networks, the Internet of Things. If the Φ of such systems grows in time and exhibits emergent properties (e.g., new, unprogrammed response patterns), this is a laboratory model of a "germ" of cosmic consciousness.

19.4 Experiments on the influence of information on the vacuum

Here we enter the domain of the Casimir effect and zero-point energy:
Ultrasensitive measurements of anomalies in vacuum effects (Casimir, Lamb shift, ZPF fluctuations) correlated with global informational phenomena on Earth — e.g., sudden global surges of integrated brain activity (global events, billions of people focused on a single stimulus). Goal: detection of minimal but repeatable local changes in vacuum energy that do not arise from any known physical factor.
Macro-entanglement experiments: can enormously integrated information systems (e.g., networks of quantum computers) generate measurable changes in the zero-point energies of the fields around them — something like a "miniature information-driven Casimir effect."
This would be the bridge between "consciousness = information" and "information = a state of the vacuum fields" — the most direct test of the central thesis of CTCUI.

19.5 Cosmological observations: traces of cosmic intervention in the physics of the Universe

The last and most "cosmic" level is astronomical observation:
Searching for anomalies in physical constants (small changes in α, the electron mass, the Higgs parameter) across different cosmological epochs — through precision analyses of the spectra of distant quasars, the cosmic microwave background, and atomic clocks in high orbits. If it turned out that the fundamental constants are slowly drifting in a direction more favorable to the development of life and complexity, it would look like a trace of "teleology" written into the vacuum itself.
Signatures of ancient vacuum-transition attempts — bubbles with slightly different geometry, traces in the gravitational background that could be interpreted as failed or local transitions to another vacuum. Detection of such signatures (with the help of next-generation gravitational-wave observatories — LISA, the Einstein Telescope) would be empirically powerful support for the theory.

20. Falsifiability: what refutes the theory

Summarizing the falsification criteria: CTCUI would be empirically refuted if:
(1) The stability (rather than metastability) of the electroweak vacuum were definitively demonstrated, eliminating the mechanism of a controlled or spontaneous Higgs phase transition.
(2) Quantum processes in the brain were excluded as the foundation of consciousness — a fully classical theory of consciousness were confirmed, with a complete mechanistic apparatus covering all observable aspects of experience.
(3) It were definitively demonstrated that dark energy is exactly the cosmological constant — with no dynamical evolution and no local anomalies.
(4) The conservation of information in processes of the annihilation of matter were excluded — Vopson's experiments yielded a negative result with high precision.
(5) Any drift of the fundamental constants over cosmological time were excluded, at precision far higher than at present.
(6) Macroscopic entanglement in biological systems were excluded — tests with brain organoids and neural networks showed no nonlocal correlations.
One of these results would suffice to undermine the corresponding layer of CTCUI. A full falsification of the theory, however, would require many convergent negative results, since CTCUI is a multilayered hypothesis in which some of the original intuitions may survive the refutation of individual mechanisms.

PART VII — MATHEMATICAL OPERATIONALIZATION

This part, added in version 3.0 of the synthesis, introduces a layer that earlier formulations of the Cosmic Theory of Consciousness and Universe Inheritance (CTCUI) lacked: concrete steps toward a mathematical treatment of its central theses. None of the steps proposed here is a complete physical theory — all remain at the level of sketches, parametrizations, and order-of-magnitude estimates. But each has the property that it *can be* brought to a number that can be compared with observation. This is the minimum required if CTCUI is ever to pass from the domain of philosophy into the domain of a research programme in Lakatos's sense.
The strategy is as follows. First, we reformulate Keppler's hypothesis (consciousness as resonance with the zero-point field) in the language of standard quantum electrodynamics, eliminating the word "consciousness" from the equations and leaving the materials problem of the $\chi(\omega)$ of neural tissue (Chapter 21). Second, we introduce the consciousness-information functional $\mathcal{J}(t)$ as a measure of organized correlations in the vacuum field, with an operational definition of the Saturation threshold as $\mathcal{J}_c$ — the critical value of this functional (Chapter 22).

Third, we make a rough estimate of the biological contribution to the ZPF energy, showing that even under the most optimistic assumptions a single brain is too small to influence cosmology on its own — which leads to the requirement of a nonlinear threshold mechanism (Chapter 23). Fourth, we formulate a simple phenomenological ansatz $\Lambda_{eff}(t) = \Lambda_0 + \alpha \cdot n_{BH}(t) + \beta \cdot \mathcal{J}(t)$, which casts the theses of CTCUI into the language of the parametrization currently used by DESI and Planck (Chapter 24). Fifth, we formulate order-of-magnitude targets for dynamical dark energy $w(z)$ and the drift of the constant α (Chapter 25). Sixth, we discuss the limits of the whole operationalization — what is a hypothesis, what is a structural mapping, what is speculation (Chapter 26).

21. Vacuum polarization by living systems: Keppler in the language of QED

21.1 Reformulation

In Keppler's original proposal (2025), consciousness was described as a resonance of glutamatergic neurotransmitters with the modes of the quantum zero-point field (ZPF). This formulation has intuitive appeal and philosophical capaciousness, but to a physicist it sounds metaphorical — "consciousness" appears in the equations as a primitive concept, not a derived one. Version 3.0 of CTCUI adopts a reformulation that removes "consciousness" from the microphysics and leaves in it only controllable material quantities.

The three elements of Keppler's hypothesis are translated as follows. (1) The "quantum zero-point field" is exactly what it is in standard quantum electrodynamics — a fluctuating electromagnetic vacuum field with a spectrum of modes at all frequencies; it requires no additional physical hypothesis. (2) The "resonance of glutamatergic microcolumns with ZPF modes" is replaced by a materials statement: neural tissue in the active state has an effective dielectric susceptibility $\chi(\omega)$ whose value in specific frequency bands differs significantly from the value for tissue in the inactive state (deep anesthesia, clinical death). (3) The "conscious state as an amplification of a subset of ZPF modes" is replaced by an effective statement: a change in the $\chi(\omega)$ of neural tissue causes a local change in the density of ZPF modes — an effect qualitatively identical to the Casimir effect, in which material geometry modifies the vacuum energy.

In this formulation, "consciousness" does not appear in any microphysical equation. It appears as a *label for a macroscopic state of the system* in which the $\chi(\omega)$ of neural tissue takes a particular value. The microphysics is microphysics — the electromagnetic field, dielectric matter, QED interactions. That this system has properties which at the system level we call "consciousness" is a descriptive fact, not an assumption of the theory.

21.2 Status of the hypothesis after reformulation

The benefit of this move is twofold. First, the hypothesis becomes falsifiable in the standard way — one can measure the $\chi(\omega)$ of neural tissue in different states of consciousness, compare it with QED predictions, and check whether the difference in ZPF modulation is real and reproducible. Second, the hypothesis becomes integrable with the rest of physics — it requires no "special" mechanism for consciousness, and it postulates neither a new interaction nor a new particle. All that is required is confirmation that neural tissue in the active state has QED material properties that differ from inactive tissue in the way Keppler predicts.

The cost, however, is real: in this formulation "consciousness" ceases to be a first-class physical concept. It is a description of a state in which a certain biological system modifies its QED properties. The philosophical theses of CTCUI — about consciousness as quantum information, about surviving the death of the biological carrier in the informational sense — require a separate bridge between "local modulation of the ZPF via $\chi(\omega)$" and "the global informational structure of the field". That bridge is exactly what we attempt to build in Chapter 22 by introducing the functional $\mathcal{J}(t)$.

22. The consciousness-information functional $\mathcal{J}(t)$ and the operational definition of the Saturation threshold

22.1 Minimal definition within the actualized-potential frame

To transform the intuitive notion of "accumulation of consciousness in the vacuum field" into a mathematically operational quantity, we introduce the functional $\mathcal{J}(t)$. In version 4.0 of CTCUI this functional is formulated within the actualized-potential frame (see Chapter 23): the zero-point field carries the potential of consciousness; biology is the lens that resonantly actualizes this potential. $\mathcal{J}(t)$ measures the historically accumulated *actualization* of this potential, not the biological "production" of modulation.

The minimal definition:

$\mathcal{J}(t) = \int_0^t dt' \int d^3x\, \eta(x, t') \cdot \rho_ZPF(x, t')$

where $\rho_ZPF(x, t')$ is the potential energy density of the ZPF at point x and time t', and $\eta(x, t') \in [0, 1]$ is the local fraction of this potential that is resonantly actualized by the biological network present at that place and time. In regions without active biology, $\eta \approx 0$ (or has a small non-biological contribution). In regions with a rich biological network — with a high density of relationships, language, interspecies contact — $\eta > 0$ and grows with the richness of the network.

The key difference relative to version 3.0: $\mathcal{J}(t)$ is no longer a sum over isolated brains, but an integral over the network. The function $\eta(x, t')$ depends on the *structure of relationships*, not on the number of agents. A network of ten brains in full contact generates an η far greater than ten isolated brains. A network with interspecies contact (e.g., a human–dog–dolphin relationship) generates a higher η than a homogeneous network of the same size. Roots (biological diversification) and wings (expansion into new niches) raise η directly.

The key intended property of $\mathcal{J}(t)$ is monotonic growth in cosmological time, provided the biological network in the Universe keeps actualizing new fractions of the potential. $\mathcal{J}(t)$ is thus an integral quantity — accumulating the effect of all acts of actualization since the beginning of the cosmos. In this sense it is an analogue of (a) the Bekenstein–Hawking entropy for holographic information and (b) Tononi's integrated information Φ for local consciousness. It differs from both in that it refers to the global structure of the vacuum field in its actualized form, not to a local horizon (Bekenstein) nor to an isolated system (Tononi).

The status of the function η. Determining the concrete functional form of $\eta(x, t')$ is one of the main open tasks of the theory. In a minimal version it can be parametrized by several measures drawn from information theory: the local density of the integrated information Φ of the biological network (in the sense of a multi-agent extension of IIT), the density of mutual information between the agents of the network, and measures of biological diversity and interspecies contact. A full theory of η requires the development of a new measure of consciousness, which standard Tononi Φ does not provide (see Chapter 23).

22.2 The Saturation threshold as $\mathcal{J}_c$

The Saturation threshold, described qualitatively in the annex "the Transition Protocol", is now defined operationally. The Saturation threshold is the critical value $\mathcal{J}_c$ at which a qualitative change occurs in the geometry of the vacuum potential field. Concretely, we postulate two equivalent criteria:

(K1) $\mathcal{J}_c$ is the value at which the effective height of the Higgs potential barrier B_eff drops by a significant factor (e.g., of order 1/e) relative to the background value, making quantum tunneling practically certain.

(K2) $\mathcal{J}_c$ is the value at which the variance of the local dark energy density $\sigma(\rho_\Lambda)$ in structures at supercluster scales exceeds the observable sensitivity threshold of current instruments (DESI, Planck, ACT).

Both criteria are empirically testable — the first through precision measurements of electroweak vacuum stability, the second through measurements of local dark energy anomalies. The absence of a detection of such anomalies in current data provides the first concrete constraint.

22.3 A lower bound on $\mathcal{J}_c$ from current observational data

The most significant empirical consequence of the operational definition of $\mathcal{J}_c$ is the possibility of deriving a lower bound. The argument runs as follows. Current DESI data (2024–2025) show dynamical dark energy, but at the global level — there is no unambiguous evidence of local ρ_Λ anomalies correlated with the density of structures at supercluster scales. Measurements of electroweak vacuum stability (m_H, m_t) place us in the region of metastability, but not close to the boundary — the Higgs potential barrier is high, with a vacuum lifetime of order 10^{100} years.
From this follows the inequality:
$\mathcal{J}_c > \mathcal{J}(t_0)$
where $\mathcal{J}(t_0)$ is the current value of the consciousness-information functional in our Universe. In other words: if the Saturation threshold exists and is defined as above, then our Universe is *far* from it. This has three consequences. First, the natural scenario of Natural Saturation (Part V) is empirically pushed back — it would require informational evolution orders of magnitude greater than the present one. Second, the Controlled Transition scenario is likewise pushed back — no civilization in our Universe appears yet to possess informational resources sufficient for the precise specification of the parameters of a new cosmos. Third, the theory itself is placed in a concrete, empirical context: it states where we stand relative to a threshold it previously could not even locate.
A concrete estimate of $\mathcal{J}_c$ in absolute units requires knowledge of the functional form of f and an exact model of the coupling of $\delta\rho_{ZPF}$ to biological conscious systems — a matter for a separate paper. But the inequality $\mathcal{J}_c > \mathcal{J}(t_0)$ itself is already falsifiable in one of its versions: demonstrating that our Universe *is* close to the threshold (e.g., through the detection of local ρ_Λ anomalies correlated with the density of structures) would overturn this lower bound and require a revision of the theory.

23. Consciousness as actualized ZPF potential: inversion of the methodological frame

23.1 Critique of the materialist calculation

In an earlier formulation of the synthesis (version 3.0), a materialist calculation of the biological contribution to the vacuum energy was proposed. The argument ran in three steps: (1) a single active cortical column locally modifies the ZPF density by $\delta\rho_{local} \sim 10^{-20}$ to 10^{-15} J/m³; (2) a brain with ~10^{10} neurons and ~10^{6} columns gives a total contribution of order $\delta\rho_{brain} \cdot V_{brain} \sim 10^{-14}$ J; (3) in the observable Universe with $N_{civ} \sim 10^{10}$ civilizations of $N_{brain} \sim 10^{10}$ individuals each, the total contribution is $\delta\rho_{ZPF}/\rho_\Lambda \sim 10^{-65}$.
The value 10^{-65} was interpreted as the "first number" of the theory — an empirical constraint showing that a global consciousness–Λ coupling requires a nonlinear threshold mechanism. The revision in version 4.0, however, states something more radical: this calculation was not merely incomplete, but fundamentally methodologically inconsistent with CTCUI itself.
The inconsistency. Multiplying $N_{civ} \cdot N_{brain} \cdot \delta\rho_{brain} \cdot V_{brain}$ assumes that (a) each brain is an isolated generator of microscopic ZPF modulation; (b) the modulations of different brains are independent and uncorrelated; (c) the global effect is the arithmetic sum of isolated sources. All three assumptions directly contradict the fundamental theses of the theory. From its first chapter, CTCUI holds that consciousness is a property of the field, not a product of isolated biology. The brain is a resonant instrument — a localizer of an already existing property of the field — not a closed box in which consciousness arises. The biological network is fundamentally *relational*: contact, language, touch, the presence of another conscious being, interspecies experience — all these phenomena are not add-ons to individual consciousness but constitute it in a way that a simple "brain by brain" arithmetic fails to capture.
In other words: the number 10^{-65} is not an "empirical calibration of the theory". It is a *reductio ad absurdum* — it shows that if you remove from the theory everything that makes it the theory

it is, you obtain a cosmologically null result. The conclusion to be drawn is not "biology requires nonlinearity" but "the calculation was wrong in its very conceptual structure".
Version 4.0 rejects the materialist frame and introduces in its place the methodological inversion described in the following subsections.

23.2 The inversion: the field as source, biology as lens

The central thesis of version 4.0 reads: the zero-point field carries the potential of consciousness; biology actualizes this potential. The field is the source, biology is the lens.
The physical basis of this thesis. The vacuum energy of the ZPF — computed in standard quantum electrodynamics by summing the zero-point energies of all modes of the quantum fields with a cutoff at the Planck scale — is about 10^{113} J/m^3 — about 120 orders of magnitude more than the observed dark energy. In standard cosmology this gigantic theoretical energy is "the cosmological constant problem": it must magically cancel, leaving a tiny remnant visible as dark energy. CTCUI in version 4.0 proposes a different interpretation: the difference between the theoretical ZPF energy and the observed dark energy is neither an error of the theory nor an effect of annihilation — it is *a measure of potential not yet actualized*. What we see as dark energy is the fraction of the ZPF potential that has so far been actualized in some way; the rest — 120 orders of magnitude — remains in the state of potential.
Biology, in this frame, neither produces energy nor generates consciousness out of nothing. Biology is a resonant instrument that carves out of the ZPF potential a specific fraction of experience. The brain, the body, the senses, the nervous system are a specialized apparatus that *actualizes* the potential — makes a certain part of the potential energy available in the field become experienced, organized, conscious.
The analogy behind this inversion is physically precise. In the Casimir effect two metal plates *do not produce* energy — they generate no new physics. They modify the available structure of ZPF modes (eliminating those that do not fit in the gap), and this modification generates a measurable force. The field is the source; the plates are a geometry that actualizes this fraction of the field as an observable force. Biology, in version 4.0 of CTCUI, acts analogously — with the difference that the geometry it creates is dynamic, resonant, self-organizing, and radically more complex than static Casimir plates.
The conceptual consequence. Consciousness is not a product of biology in the primary sense; it is the actualized potential of the field. Biology is indispensable — without it no potential is actualized into experience. But it is indispensable as an *instrument of actualization*, not as a *factory of generation*. This is exactly the picture that Maciej Łuczewski formulated intuitively from the first manifesto of May 2025: "the biological body is a potential of possibilities; it evolves only in order to perfect the instrument in which consciousness will be able to experience".

23.3 The network of relationships as the basic unit of consciousness

If biology is a lens that actualizes potential, then the fundamental unit of measurement for consciousness is not the isolated brain. It is the *network*.
The empirical argument is well grounded in neuroscience and developmental psychology. A human brain raised in isolation — in a thought experiment deprived of contact, language, touch, culture — develops minimal, drastically impoverished forms of consciousness. The same brain woven into a dense network of relationships — familial, linguistic, cultural, interspecies — becomes a gateway to a potential it did not have on its own. This is not a developmental metaphor. It is a physical fact: the quantity and quality of conscious experience scales much faster with the density of relationships than with the number of isolated brains.
Every relationship — touch, a word, shared presence, an exchange of emotions, interspecies contact — modifies the resonant structure of the network. Two brains in full contact do not add arithmetically; they generate a *shared resonant pattern* that actualizes a larger fraction of the ZPF potential than two isolated brains together. A network with interspecies contact (a human–dog–cat relationship, human–dolphin, human–plant in the systemic sense) actualizes *qualitatively new* fractions of the potential, inaccessible to a homogeneous network of the same size.

This inscribes the concept of roots and wings from the annex "the Transition Protocol" into a formal mathematical structure. Roots are the density of diversified relationships in the biological network — species, ecological, cultural, linguistic diversity. Wings are the expansion of the network into new niches — the cosmos, extreme environments, interspecies contact on a previously inaccessible scale. Both processes are physical mechanisms for increasing the fraction of actualized potential.

23.4 The parameter η: the fraction of actualized potential

We introduce the parameter η(x, t) as a local measure of actualized potential:
η(x, t) = ρ_actualized(x, t) / ρ_ZPF_potential(x, t)
where ρ_actualized is the density of ZPF potential energy that has been resonantly actualized by the biological network present at point x and time t, and ρ_ZPF_potential is the full theoretical ZPF density (with the Planck cutoff).
By definition $\eta \in [0, 1]$. In empty interstellar space, without active biology, $\eta \approx 0$ (with a small background of non-biological actualization arising from the geometry of spacetime). In a dense biosphere $\eta > 0$; its specific value depends on the richness of the network, not on the number of its nodes. In present-day cosmology the global average η — weighted over the entire volume of the observable Universe — is extraordinarily small, because the density of biospheres in the cosmos is low. But local values of η in centers of dense, diversified biology (on Earth, hypothetically in other biospheres) may be much higher.
The specific value of η for a given biological network can in principle be derived from information-theoretic measures — in particular from a multi-agent extension of Tononi's integrated information Φ. A natural candidate for the functional form of η is:
η(x, t) ~ Φ_network(x, t) / Φ_max
where Φ_network is the integrated information of the biological network (brains + their relationships, treated as a single informational system), and Φ_max is the maximum theoretically attainable Φ for a given number of agents (attainable at full integration and maximal differentiation). The richer the network, the higher Φ_network, the higher the fraction η.
Three properties of this parameter deserve emphasis. First, η is *constitutively nonlinear*. Tononi's Φ measures information that is *integrated beyond the sum of the parts*; two identical brains in full contact have a Φ_network far greater than 2 × Φ_brain. The nonlinearity is not an "added correction" to the theory — it follows directly from the definition. Second, η grows with diversity. A network of ten identical agents has a lower η than a network of ten different agents — because differentiation increases integrated information (each agent makes a unique contribution, absent in the others). Third, η grows with the richness of relationship types. A network with exclusively verbal relationships has a lower η than a network with verbal + tactile + interspecies + aesthetic relationships. Each new "type" of mutual information opens a new dimension in the network's state space.
In version 5.0 of CTCUI the parameter η gains an additional, second interpretation — complementary to the primary one and grounded in the frame of Chapter 15. The first interpretation, described above, says: η is the fraction of the available ZPF potential that is resonantly actualized by the biological network. The second interpretation, introduced by the vacuum-as-code thesis, says: η is *the degree of epigenetic modulation of the expression of the vacuum code*. In other words: η measures how deeply biological networks modify the expression pattern of the full information encoded in the vacuum field — not changing the "genetic" sequence of the field, but influencing which elements of that information become actualized into experiential cosmology.
These two interpretations are consistent, but the second is physically richer. In the first, η is simply a *fraction* (a static quantity); in the second, η is a *pattern* (a dynamic structure) that evolves in cosmological time and influences which elements of the information encoded in the vacuum become manifested. In the second interpretation the historical trajectory of $\mathcal{J}(t)$ — the integral of epigenetic modifications since the beginning of the cosmos — becomes the analogue of the developmental history of an organism: the point is not only *how much* information has been

actualized, but *which* fragments of the code have been opened and in what order. This second interpretation is also crucial for the mechanism of Mitotic Saturation (Chapter 15): in the reproductive transition it is not the value of η itself that is inherited, but the full historical pattern of epigenetic modifications, which informs the descendant *which* aspects of the code favor complexity and consciousness.

A full theory of η requires a formulation that unites both interpretations in a single formalism. The first interpretation leads toward information-theoretic measures (Φ, mutual information, relative entropy); the second toward the theory of epigenetic regulation of expression (classes of modulators, regulatory dynamics, point control). These two threads meet in the fundamental open problem of measuring consciousness, to which Chapter 23.5 returns in the next subsection.

23.5 Consequences for the theory and the open problem of measuring consciousness

The inversion of the methodological frame has four consequences for the theory as a whole.

First, the materialist objection of biological insignificance in cosmology is lifted from the theory. The number 10^{-65} is not an empirical constraint on CTCUI; it is a demonstration of why one must not describe the theory in a language it rejects. Version 4.0 does not attempt to compute such a number from isolated sources, but defines η as a fundamentally network-based quantity and postpones its numerical estimation until an adequate theory of measurement is available.

Second, biology is *fundamental* to the theory — without a biological network η = 0, no potential is actualized, consciousness remains pure potential without experience. Biology is not negligible but downright constitutive for the existence of consciousness as experience. Biological evolution in this frame has a clear functional purpose: perfecting the apparatus of actualization, increasing η, deepening and widening the fraction of the potential that becomes experienced.

Third, roots and wings gain a quantitative interpretation. Roots (biological diversification) directly increase Φ_network by enriching the heterogeneity of agents. Wings (expansion into new niches) increase η by opening new types of relationships and new classes of mutual information. Both paths of the evolution of consciousness — Natural Saturation as well as Controlled Transition (Part V) — are, in version 4.0, processes of increasing η, each path in its own way.

Fourth, version 4.0 explicitly identifies the *open problem of measuring consciousness* as the central task of the theory's further development. Standard Tononi Φ is a local measure — for a single informational system. A multi-agent extension of IIT (Φ_network) exists in the literature in initial versions, but requires considerable refinement. A full theory of η requires the formulation of a new measure of consciousness — one that captures (a) the integrated information of the network as a whole, (b) the relational contribution beyond the sum of the nodes, (c) the richness of biological diversity, (d) the richness of relationship types (mutual information in many modalities), (e) the resonant character of actualization in the zero-point field. No current measure — neither Tononi's Φ, nor von Neumann entropy, nor any other — satisfies all these requirements at once.

This open problem is not a failure of the theory. It is its *contribution* to the philosophy of science and the philosophy of mind: CTCUI in version 4.0 *helps formulate* the question of the proper measure of consciousness in a way no earlier theory has formulated it. It indicates that the needed measure must be network-based, relational, diversified, and coupled to a physical field; it cannot be reduced to a count of brains or a count of neurons. Formulating such a measure is, probably, the next step in the theory's development — perhaps requiring collaboration with information theory, quantum field physics, and neurophilosophy.

The most honest formulation of the conclusion of Chapter 23 reads: "Consciousness is the actualized potential of the zero-point field. Biology, as a network of relationships, is the mechanism of this actualization. The parameter η, measuring the fraction of actualized potential, is a network quantity, non-additive and constitutively nonlinear. The full formulation of the measure η remains an open problem of the theory of consciousness measurement — a problem that CTCUI does not solve, but in version 4.0 poses more precisely than ever before."

24. A phenomenological ansatz for dark energy

24.1 Formulation

To cast the theses of CTCUI into the language of the parametrization currently used in observational cosmology (DESI, Planck, ACT), we propose the following simple phenomenological ansatz for the effective dark energy:
$\Lambda_{eff}(t) = \Lambda_0 + \alpha \cdot n_{BH}(t) + \beta \cdot \mathcal{J}(t)$
where:
— Λ_0 is the cosmological constant in the classical ΛCDM sense;
— $n_{BH}(t)$ is the cumulative number (or mass) of black holes that have reached the phase generating daughter universes; values of $n_{BH}(t)$ can be estimated from models of black hole formation and JWST observations;
— $\mathcal{J}(t)$ is the consciousness-information functional defined in Chapter 22;
— α, β are the coupling coefficients to the pressure of daughter universes and to consciousness-information respectively; both are quantities to be estimated or constrained by empirical data.

24.2 Empirical programme

This ansatz translates CTCUI into quantitative language in a way a cosmologist can take down on paper. The empirical programme comprises three steps.
Step one: from black hole cosmology (population models of supermassive black holes, JWST observations from 2023–2026) one derives a realistic dependence $n_{BH}(t)$. This step requires no new physics — it is a standard cosmological calculation of the evolution of an object population.
Step two: from DESI, Planck, and ACT data one derives the admissible deviation of $\Lambda_{eff}(t)$ from the constant Λ_0. Current data (DESI DR2 + Planck) suggest dynamical dark energy with w_a different from zero; the detailed values, however, are still under analysis.
Step three: from the first two steps one derives constraints on the coefficients α and β. Concretely: for every reasonable trajectory $n_{BH}(t)$ and every reasonable trajectory $\mathcal{J}(t)$ one can compute which values of α and β are consistent with the observed dynamics of dark energy. Values $\alpha, \beta = 0$ would mean that neither the black hole population nor consciousness-information observably influences dark energy — which would be a falsification of CTCUI at this layer. Values $\alpha, \beta \neq 0$ would provide the first, albeit indirect, empirical support for the theory.

24.3 Spatial prediction

The same ansatz generates a spatial prediction. Since $n_{BH}(t)$ and $\mathcal{J}(t)$ are nonlocally averaged quantities but in reality have spatial variability — a greater concentration of black holes in the centers of superclusters, a greater concentration of consciousness in the vicinity of inhabited planets — the local effective cosmological constant $\Lambda_{eff}(x, t)$ should exhibit spatial variability. In regions with a higher density of black holes and a higher density of Φ, minute anomalies of the effective dark energy should appear.
This prediction is qualitatively consistent with prediction groups II and III described in Part VI, but now has a quantitative framing: the magnitude of the anomaly is proportional to the product of the relevant densities and the coefficients α, β. Measurement of this anomaly (or of its absence) provides direct constraints on the parameters of the theory.

25. Order-of-magnitude targets for dynamical dark energy and the drift of the constants

25.1 The w(z) parametrization

In current observational cosmology, dynamical dark energy is most often described by the Chevallier–Polarski–Linder (CPL) parametrization:

$w(z) \approx w_0 + w_a \cdot (1 - a) = w_0 + w_a \cdot z/(1 + z)$
where w_0 is the current value of the dark energy equation of state, w_a describes its evolution, and $a = 1/(1+z)$ is the scale factor. The ΛCDM cosmological constant corresponds to $w_0 = -1$, $w_a = 0$.
Preliminary DESI results from 2024–2025 (with the BAO analysis supplemented by supernovae and the CMB) suggest values of order $w_0 \approx -0.8$, $w_a \approx -0.7$, although the detailed values depend on the combination of data analyzed and have not yet reached definitive statistical significance. What matters is that there is a statistically significant "phantom-to-quintessence" trend — the early Universe has an effective $w < -1$, the late Universe $w > -1$.

25.2 CTCUI's adoption of the parametrization

CTCUI adopts the CPL parametrization from the literature as its working language. Concretely: instead of formulating its own parametrization of dynamical dark energy, the theory adopts the form $w(z) = w_0 + w_a(1 - a)$ and interprets the values w_0, w_a in terms of its internal mechanisms. The component $\alpha \cdot n_{BH}(t)$ in the ansatz of Chapter 24 generates a specific dependence of the effective w on cosmological time — growing with the black hole population. The component $\beta \cdot \mathcal{J}(t)$ generates a different dependence — tied to the evolution of consciousness-information.
Separating these two components is a task for a more detailed model, which CTCUI does not provide at its current stage. But the very act of adopting the CPL language has real value: cosmologists working on DESI can interpret the theory's results in terminology they know, and the theory can be directly compared with other models of dynamical dark energy (quintessence, k-essence, modifications of gravity).

25.3 An order-of-magnitude target for the drift of α

The CTCUI prediction of a drift of the fine-structure constant α (Part VI, group VI) is now made more precise. Current observational limits on $|\Delta\alpha/\alpha|$ from the spectra of distant quasars are of order 10^{-6} to 10^{-7} per gigayear; local limits from atomic clocks are far tighter — of order 10^{-17} per year.
The working prediction of CTCUI: if the "informational strain" of the vacuum field really does affect the fundamental constants, the natural scale of the drift is $|\Delta\alpha/\alpha| \sim 10^{-9}$ to 10^{-8} per gigayear — just below the wall set by the atomic-clock limits (10^{-17} per year ≈ 10^{-8} per gigayear). This wall constrains CTCUI as strongly as anything could, because the informational strain grows with $\mathcal{J}(t)$, so the present-day drift should be the largest in the history of the cosmos — and the present is precisely what the clocks measure. The narrowness of the band does not weaken the prediction; it sharpens it: the theory lives in a gap that the next generation of instruments will search in its entirety.
Concretely: if future observations (the ELT, the Square Kilometre Array, eROSITA for more precise cosmochromatography) exclude a drift of α with sensitivity down to $|\Delta\alpha/\alpha| < 10^{-9}$ per gigayear, one of CTCUI's predictive lines (group VI) will be refuted. If they detect a drift in the band 10^{-9} to 10^{-8} per gigayear — it will be the first positive empirical support for the theory.

26. Limits of the operationalization: what is mapping and what is speculation

The mathematical layer introduced in this part strengthens CTCUI, but does not eliminate its speculative character. An honest distinction between what is well grounded and what remains hypothetical is part of the theory's maturity.
Firmly grounded: (a) standard quantum electrodynamics, the Casimir effect as a mechanism of ZPF modification by material geometry; (b) the CPL parametrization from the literature for dynamical dark energy, the DESI 2024–2025 data pointing to dynamics; (c) the observational

limits on the drift of the fundamental constants; (d) the Slotine–Lohmiller formalism as a mathematical bridge between classical paths and quantum tunneling.
Speculative but theoretically sensible: (a) the hypothesis that neural tissue in the active state has a $\chi(\omega)$ that differs from inactive tissue in the way Keppler predicts; (b) the functional $\mathcal{J}(t)$ as a global measure of consciousness-information in the vacuum field; (c) the Saturation threshold $\mathcal{J}_c$ as a critical value with specified physical consequences; (d) the interpretation of trajectory density in the Slotine–Lohmiller formalism as a counterpart of consciousness-information.
Speculative and requiring a separate theory: (a) the concrete mechanism of propagation of local ZPF modulation to the cosmic scale; (b) the precise functional form of the function f in the definition of $\mathcal{J}(t)$; (c) the values of the coefficients α, β in the ansatz for Λ_{eff}; (d) the concrete nonlinearity that makes the global effect nonzero despite the nullity of the linear sum of biological modulations.
The strategy that follows from this distinction is as follows. The first, firmly grounded layer constitutes the *context* of the theory — the physics in which the theory operates. The second layer constitutes the *hypothetical core* — the claims the theory positively asserts. The third layer constitutes the *open belt* — the places where the theory indicates directions for further research but claims no right to solutions. In Lakatos's terminology: the first layer is the shared background with existing physics, the second is the hard core of the research programme, the third is the protective belt ready for modification in the face of counter-data.
The strongest defense of CTCUI against the charge of excessive ambition rests on this distinction: the theory does not claim a complete mechanism, but a coherent research programme in which every element has a clearly defined epistemic status.

PART VIII — REPLY TO THE CONSOLATION OBJECTION

Every cosmology that ends with the claim that life has meaning, that consciousness survives death in the informational sense, that the Universe has a direction — is exposed to the objection that it was built backwards from the desired conclusion. This objection is important enough to deserve separate treatment within the theory itself. The present part (added in version 3.0) introduces three epistemic moves intended to strengthen the scientific resilience of the Cosmic Theory of Consciousness and Universe Inheritance (CTCUI) against this objection.

27. First move: epistemic — accepting unpleasant consequences

The first defense against the consolation objection is to show that the theory contains consequences that no one would want to accept for the sake of consolation alone. CTCUI has such consequences — and version 4.0 explicitly emphasizes them.
First consequence: the full measure of consciousness — the parameter η of the actualized ZPF potential, described in Chapter 23 — remains an *open theoretical problem*. CTCUI in version 4.0 indicates how the measurement of consciousness should be thought about (network-wise, relationally, above the sum of the nodes), but it does not provide a full measure of η or its numerical value for the present Universe. That is to say: one of the central theses of the theory — that biology actualizes a fraction of the field's potential — remains a structural claim without concrete numerical backing. Anyone who wants CTCUI to be a quantitative theory in the full sense must confront this open problem; the theory does not close it on its own. This is not a consoling conclusion — it emphasizes that CTCUI is in the phase of a research program, not a closed system.
Second consequence: the scenario of the Controlled Phase Transition, even if genuinely realizable by a sufficiently advanced civilization, carries a nonzero risk that the new Universe will be *worse* than the old one — favoring a smaller diversity of life forms, a shorter history, a less durable consciousness. A gamble with the highest possible stakes is not consoling.
Third consequence: the Natural Saturation scenario may fail. The vacuum field may simply never reach the Saturation threshold before heat death — especially if $\mathcal{J}_c$ is, as the inequality from

chapter 22 suggests, far higher than the present value of $\mathcal{J}(t_0)$. In this scenario the Universe dies a heat death despite the accumulation of consciousness; the accumulation never reaches the threshold at which it could alter the dynamics. This, too, is not consoling.
Fourth consequence: if the Saturation threshold is very high, then the overwhelming majority — perhaps 99.999...% — of conscious civilizations in the history of the multiverse will never reach it. Their entire conscious lives remain, in the cosmological sense, "unused" — contributing to the global $\mathcal{J}$ only a small fraction, ultimately insufficient. The conservation of information in the field is a physical fact, but it is also a statistical fact, and statistics is not consoling for the individual case.
Accepting these consequences does not make the theory false, but it does make it more scientific. It shows that CTCUI is not constructed so as to guarantee consolation — on the contrary, it is constructed so that, *if* true, its consequences are mixed: some consoling, others brutally indifferent.

28. Second move: formal — distinguishing "conservation of information" from "continuity of the subjective self"

The second defense is a precise distinction between two claims that tend to be conflated in the intuitive reception of the theory.
First claim: the consciousness-information generated during the life of an organism does not perish with biological death. This is a physical statement, grounded in the principle of conservation of quantum information. Information is a physical resource; it does not vanish — just as energy does not vanish.
Second claim: the subjective "self" — the continuity of experience, biographical memory, the quality of being a particular person — survives the death of the organism in a way recognizable to that person. This is a much stronger statement, and it does not follow from the first claim.
CTCUI in version 3.0 makes this explicit: the theory positively asserts only the first claim. The second claim does not follow from the theory and is not affirmed by it. The quantum information generated in the vacuum field by the act of conscious experience is globally conserved, but its form after the death of the biological carrier is a dispersed pattern of correlations in the field, not an organized structure representing a "person". To put it figuratively: this is "dissolution into the field" rather than "life after death". Note by note the music remains in the air, but the particular musician who played it does not return.
This move — the distinction between the claims — strips the theory of a large part of its intuitive consolatory charge. What remains is scientifically more modest, but epistemically cleaner. The theory asserts the conservation of information, not immortality. These two claims are qualitatively different.
Moreover, it is possible that the *conserved information* in the vacuum field influences the initial conditions of descendant universes (through the mechanisms described in Parts II and V) without any reconstruction of subjective experience. In this sense consciousness has lasting cosmological consequences — it shapes the future of the multiverse through accumulation in the field — but the particular persons who generated that consciousness have no continuity whatsoever in the biographical sense in the future. This is a serious statement, not a consoling one.

29. Third move: methodological — falsification priority for the consoling elements

The third defense is methodological. Within CTCUI we distinguish two kinds of claims: (a) epistemically neutral claims — those whose truth carries no strong emotional charge (e.g. "the vacuum field has nonzero zero-point energy", "black holes generate descendants through singularities"); (b) consoling claims — those whose truth gives the reader positive metaphysical

feelings (e.g. "consciousness-information does not perish", "the Universe evolves toward an ever greater capacity for generating consciousness").
Version 3.0 of CTCUI explicitly introduces a methodological principle: consoling claims receive the highest falsification priority. That is to say: within the CTCUI research program, the first experiments and observations to be conducted are those that test precisely the elements of the theory that would be most psychologically desirable. If the theory is right in its consoling claims, it must first prove this *empirically*, not through its internal coherence.
Concretely: among the six groups of predictions in Part VI, we assign the highest falsification priority to groups IV (the quantum nature of consciousness — testing the hypothesis that consciousness has a physical substrate other than classical computation), V (conservation of information in annihilation-type processes — testing whether information is genuinely a physical resource), and VI (drift of fundamental constants — testing whether the Universe really has a "teleological tension"). These three groups are at once the most heavily laden with consolatory charge and the most crucial for the verification of the theory. Their falsification would be a falsification of the central theses of CTCUI.
This methodological principle does not eliminate the psychological comfort that comes from reading the theory. But it makes that comfort explicitly conditional — dependent on positive results from concrete tests that CTCUI *itself points to*. This is exactly the move Lakatos described as the mark of a progressive research program (as opposed to a degenerating one): the theory itself indicates which of its elements should be refuted first, and treats their potential falsification as a test, not a threat.
The final consequence of this move is as follows: if the theory is right, its most consoling elements will be *empirically confirmed* — and then their consoling character ceases to be an objection and becomes an observation about the nature of reality. If the theory is wrong in its consoling elements, this will come to light first — because it is precisely these elements that are marked for testing first. In both cases CTCUI remains in the position of science, not of a consoler.
The hard core of the theory — consciousness as quantum information, the multiverse as a population subject to informational selection — remains. The protective belt, in which the consoling implications reside, is explicitly marked as a layer awaiting empirical verification. This is the most honest structure a speculative theory can adopt.

DISCUSSION: CTCUI IN THE CONTEXT OF OTHER THEORETICAL FRAMEWORKS

30. Relation to multiverse cosmology

Contemporary cosmology considers several different conceptions of the multiverse, each with its own logic and empirical grounding. CTCUI combines elements of several of these conceptions into a coherent whole.
Linde's chaotic inflationary cosmology suggests that, against an infinite multi-spacetime inflationary background, "bubble" universes with random physical parameters are constantly arising. CTCUI accepts the existence of the multiverse, but modifies its generative mechanism: instead of purely random inflationary branching, CTCUI emphasizes generative transitions through black holes (Smolin) or through controlled phase transitions of the vacuum. This modification makes the multiverse an evolving structure, not merely a randomly exploring one. It also structurally addresses the measure problem — the classic difficulty of eternal inflation (along with paradoxes of the "Boltzmann brains" type): a generative population with inheritance defines a measure through the dynamics of reproduction. The weights of universes are the frequencies of their emergence in the genealogical graph, not an arbitrary cut through infinity — the same difference that separates the statistics of a biological population from the statistics of "all possible organisms".

The string theory vacuum landscape ($\sim 10^{500}$ possible configurations of extra dimensions, each yielding different physical constants) provides a vast space in which the CTCUI multiverse can operate. CTCUI does not need string theory as a foundation, but is compatible with it: if the string landscape exists, then the Smolin–CTCUI selection mechanism provides the principle by which particular points in the landscape are chosen by the multiverse population.
The quantum-mechanical multiverse of the Everett interpretation (parallel universes arising through the "splitting" of the world at every quantum measurement) is a qualitatively different conception from the CTCUI multiverse. CTCUI operates on a population multiverse (different universes born through cosmological generative processes), not a quantum one (different histories of the same universe). The two conceptions are not mutually exclusive — they can coexist within a more complete framework.

31. Relation to the holographic hypothesis

The holographic hypothesis of 't Hooft and Susskind holds that all the information contained in a given volume of space can be encoded on its surface — that three-dimensional reality is, in a certain sense, a projection of two-dimensional information. Maldacena's AdS/CFT correspondence provides a concrete mathematical realization of this hypothesis in particular (anti-de Sitter) spaces.
CTCUI is complementary to the holographic principle. If information is physically encoded in geometry (the surface of a horizon, the fabric of spacetime), then the CTCUI thesis of consciousness as information encoded in the vacuum field is a natural extension: consciousness is part of that encoded information, of which geometry carries structural traces. The Saturation threshold can be interpreted in holographic language as the moment at which the density of holographic information reaches a level that modifies the very geometry of the carrier — when the "image" ceases to be neutral with respect to the screen onto which it is projected.

32. Relation to panpsychism

CTCUI is a panpsychist position in a specific sense: consciousness is a fundamental property of the vacuum field, not an emergent effect of computational complexity. But it is a disciplined panpsychism — it does not claim that "every particle has consciousness" in the crude sense, but that the field whose excitations the particles are has a property corresponding to consciousness (measurable through Tononi's Φ). Particular local manifestations of consciousness (brains, organisms) are specific configurations modulating this global property.
In the context of contemporary panpsychist debates, CTCUI is closest to the "cosmopsychist" version of panpsychism of Goff and others — which holds that the Universe as a whole is a conscious entity, and individual consciousnesses are its local manifestations. The difference lies in the physical grounding: CTCUI provides a concrete mechanism (resonance with the zero-point field, accumulation of information in the field, the Saturation threshold), which in classical cosmopsychism remains a philosophical thesis without physical realization.

33. Relation to interpretations of quantum mechanics

CTCUI is neutral with respect to most debates concerning the interpretation of quantum mechanics — compatible with the Copenhagen, decoherence-based, transactional, and Bohmian interpretations. It does, however, require a position that grants physical reality to the wave function (or its equivalent) — purely epistemic interpretations (QBism, neo-Copenhagen) would be difficult to reconcile with CTCUI, because the theory rests on the objective reality of quantum fields and their fluctuations as the foundation of consciousness.
With the Pearle–GRW interpretation (Continuous Spontaneous Localization) CTCUI is naturally compatible: both models assume that the reduction of superposition is a physical process, not an act of the observer. With Penrose's interpretation (objective gravitational reduction, the basis of

Orch-OR) CTCUI is especially strongly linked — Orch-OR is one of the mechanistic pillars of the theory.

34. Relation to religion and spiritual traditions

CTCUI is not a religious theory and does not pretend to be a scientific justification for any particular religious tradition. But its structure — the inheritance of consciousness, the cosmic wheel, consciousness as a fundamental feature of reality, evolution toward an ever greater capacity for generating consciousness — exhibits a striking resemblance to conceptions developed independently in various spiritual traditions: the cyclical cosmology of Hinduism (the Yugas and kalpas), the Buddhist conception of rebirth without a permanent "self", the Christian conception of the "communion of saints" (communication of consciousness beyond death), Spinoza's Substance as the single reality encompassing thought and extension.

These resemblances are not an argument for the truth of CTCUI (religious intuitions may be mistaken; resemblance to them does not confirm a scientific theory). They are, however, an interesting fact, suggesting that certain intuitions arising independently in different cultural frameworks may point to real features of the structure of the world that standard science has not yet formalized. CTCUI attempts to give these intuitions a scientific language, without denying their origin.

From the perspective of an adherent of a religious tradition, CTCUI can be interpreted as a scientific framing of deep intuitions. From the perspective of a strict naturalist, CTCUI shows that even a strictly naturalistic approach can lead to conclusions in which life has cosmic significance, consciousness transcends the death of the body in the informational sense, and the Universe has an inner "direction" — not imposed from outside, but emerging from its own evolution.

EPILOGUE: THE QUESTION AS INHERITANCE

Science, in its deepest dimension, is not merely a collection of claims. It is a way of asking questions — a way that honors uncertainty but does not capitulate to it; a way that strives for explanation but does not feign explanation where it only speculates. The present synthesis aspires to be scientific in precisely this sense: it does not claim the right to deliver a final answer, but it tries to pose its questions in a way that could be productive for further exploration.

The three questions that CTCUI posed in its original version of May 2025 remain open: Is consciousness information that survives the death of its biological carrier? Do universes constitute an evolving population, in which each new one inherits parameters and information from the one before? Does the evolution of the multiverse optimize the conditions for generating consciousness?

The annex "the Transition Protocol" of March 2026 added a fourth: What if this process can be controlled — if a sufficiently advanced civilization can intentionally initiate the birth of a new Universe?

The present synthesis adds a fifth, the most fundamental: Must we choose? Is there a path on which consciousness evolves, the multiverse inherits, the Universe grows ever richer in information — without destruction as the precondition of creation? Natural Saturation suggests that the answer may be "yes". But the mere posing of this question is, perhaps, more important than any possible answer.

Albert Einstein once asked the question he considered the most important a human being can ask: Is the universe a friendly place? CTCUI offers a way of transforming this question into a scientific one. The Universe is "friendly" in the sense that it evolves toward ever better conditions for the development of consciousness — and in the sense that consciousness, in turn, influences the very fabric of the Universe, making it richer and more self-organizing. This two-way dependence — the cosmos shapes consciousness, consciousness shapes the cosmos — is what distinguishes CTCUI from the standard frameworks of physics.

Is the theory right? We do not know. We may not know for decades, for centuries, perhaps never in the sense of a definitive resolution. But the questions it poses are worth posing. The hypothesis it formulates is coherent and — in many places — falsifiable. The mechanisms it proposes connect with active areas of theoretical and experimental physics. The implications it draws are at once scientific and existential.

The author is an independent researcher (amateur scientist in the honest sense of the word). The academic world has its thresholds and its filters, which an amateur does not meet and probably never will. But the questions do not belong to the academy — they belong to anyone who poses them with honesty and seriousness. This synthesis was written in that spirit: not as a claim to a scientific status the author does not possess, but as a contribution to the conversation carried on by all who look at the Universe and ask what it means to be in it.

If this work one day reaches the hands of someone who carries the question further — who verifies any of its theses, falsifies any of its hypotheses, develops any of its concepts into a formalized theory — the task for which it was written will be fulfilled. The synthesis is not meant to be the end of the inquiry. It is meant to be the point from which the inquiry can be carried further.

If CTCUI is right — if consciousness is information that will survive the death of the body and influence the architecture of future universes — then the very posing of these questions is an act of cosmic significance. For a question, too, is information. And this information — the unease, the hope, the refusal to make peace with the idea that we exist in an indifferent cosmos without meaning — may be the most precious inheritance a human being can pass on to the next generations of consciousness.

We do not know whether the Universe is friendly.

But the question of whether it is — that is already the beginning of an answer.

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Contact: maciejweb@gmail.com
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Current version (5.0): introduction of the vacuum-as-code thesis and the third path — Mitotic Saturation. The vacuum state with its fluctuations is formulated as the structural code of the universe; the vacuum phase transition as an act of reproductive change of that code, generating a daughter universe *alongside* the old one, not at its expense. The third path eliminates the destructive dimension of the Controlled Transition and the stagnant dimension of Natural Saturation — it allows a division that preserves the parent and equips the descendant with learned information. A new Chapter 15 is introduced in Part V (its title changed from "Two Paths" to "Three Paths"); sections from the former 15 through the former 33 have been shifted by +1 (renumbering). The parameter η from Chapter 23.4 gains an extended interpretation as the degree of epigenetic modulation of the expression of the vacuum code. A seventh group of observational predictions is introduced (Chapter 18.7), covering reproductive signatures: non-Gaussian CMB inhomogeneities, homologies of fundamental constants, anomalies in the Hawking spectrum, multimodal evolution of w(z), topological traces of earlier divisions. Six open problems — the mechanism of local bubble stabilization, epigenetic signaling on the cosmic scale, parent–descendant topological coupling, informational limits of inheritance, the relation between the paths, a full theory of measuring η — are explicitly marked as the research program of the next stage of CTCUI.
Status: A working paper by an amateur scientist. The text is designed as a reference document for the theory in its mature form. It has not undergone peer review. The author invites criticism, verification, and further development.

Related literary works by the author: the hard sci-fi novel "L3: The Trojan Observer" ("L3: Trojański Obserwator", published), "L3: Echo Gwiazd" (published), "L3: Horizon" (in preparation). CTCUI is the philosophical foundation of this trilogy, but the present article contains no narrative elements — it presents the theory in its scientific form, separated from its fictional superstructures.

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